



A novel vector-field-based motion planning algorithm for 3D nonholonomic robots[☆]

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ABSTRACT

This paper focuses on the motion planning for mobile robots in 3D, which are modeled by 6-DOF rigid body systems with nonholonomic kinematics constraints. We not only specify the target position, but also impose the requirement of the heading direction at the terminal time, which gives rise to a new and more challenging 3D motion planning problem. The proposed planning algorithm involves a novel velocity vector field (VF) over the workspace, and by following the VF, the robot can be navigated to the destination with the specified heading direction. In order to circumvent potential collisions with obstacles and other robots, a composite VF is designed by composing the navigation VF and an additional VF tangential to the boundary of the dangerous area. Moreover, we propose a priority-based algorithm to deal with the motion coupling issue among multiple robots. Finally, simulations are conducted to verify the theoretical results.

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1. Introduction

As a fundamental problem in robotics and control, motion planning refers to finding a path or trajectory which guides robots from an initial position to a goal position, without collisions with obstacles or other robots. Motion planning algorithms have been employed in a wide range of robotic systems, e.g., planar mobile robots (Li et al., 2021b; Zhao & Zhu, 2022), marine vessels (Hassani & Lande, 2018; Zhou et al., 2022), and quadrotors (Tordesillas & How, 2022; Zhou et al., 2021). Generally, to reduce weight and cost, robotic systems are often designed with fewer actuators than the degrees of freedom (DOFs), resulting in underactuated systems. From the kinematic level of mobile robots in 3D, a typical form of underactuation is that the robot modeled by a 6-DOF rigid body has no sway and heave velocity, while only one

surge velocity and three angular velocities serve as control inputs. Such a system model is referred to as a 3D nonholonomic rigid body, and can be utilized to describe the kinematics of various robotic systems, such as fixed-wing UAVs (Fari et al., 2020; He et al., 2024) and autonomous underwater vehicles (He et al., 2022; Li & Wang, 2013).

The motion model of a 3D nonholonomic mobile robot can be described by a nonholonomic constrained rigid body with six DOFs. Resulting from the nonholonomic constraints, the heading direction of the robot is restricted and always points to the x-axis of the body-fixed frame. Motivated by this fact, we intend to take into account not only the goal position but also the heading direction of the robot in motion planning. For instance, in real-world scenarios, the heading directions of multiple missiles are typically specified in the terminal guidance so as to realize a better performance of coordinated attack. Similarly, in surveillance tasks performed by multi-UAV systems, the final orientation of each UAV should point to a certain direction to obtain the largest overall surveillance area. Therefore, from the perspectives of theory and practice, it is of great importance to investigate the motion planning problem with a specified position and a heading direction.

However, it is not trivial to simultaneously plan the position and heading direction of 3D nonholonomic robots, and the main challenge arises from the underactuation caused by the nonholonomic constraints. On one hand, since the linear velocity is

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restricted to the heading direction, we should adjust the heading direction towards the destination such that the robot can move towards the goal position. On the other hand, the heading direction should be employed for reaching the destination, while it has to satisfy the requirement of terminal direction, which demonstrates the underactuated characteristics of 3D nonholonomic robots. Besides, compared to the 2D case which has only one rotational DOF, the rotation control in 3D is more complicated due to the coupling among three different rotational DOFs, leading to a more challenging motion planning problem.

There exist several commonly-used methodologies for motion planning, including the roadmap method (Bhattacharya & Gavrilova, 2008; Lehner & Albu-Schaffer, 2018), the cell decomposition approach (Cai & Ferrari, 2009; Cowlagi & Tsiotras, 2012), and the sampling-based algorithm (Karaman & Frazzoli, 2011; Oh et al., 2021). These methods generally do not consider the robot dynamics or regard the robot as a point of mass, which thus cannot be directly applied to the nonholonomic systems. Although some optimization-based algorithms can handle the nonholonomic constraints, such as Bloch et al. (2021), Cichella et al. (2021), Li et al. (2021a), Zhao and Zhu (2021), the feasibility of the optimization problem cannot be guaranteed or it suffers from heavy computational burden once the nonholonomic constraints are taken into consideration. Besides, the control inputs derived from these optimization-based algorithms are open-loop, relying only on time, thereby not robust to disturbances.

In addition to the above planning methods, several papers investigate how to simultaneously regulate the position and orientation (i.e., the configuration) of a 3D nonholonomic robot by control techniques, such as Aicardi et al. (2001), Egeland et al. (1996), Li et al. (2018), Mirzaei et al. (2016). These papers treat the robot as a control system with states consisting of both position and orientation, and then stabilize such a system by using the Lyapunov stability theory. Although the states of the robot can be regulated to specific values, these studies focus only on the obstacle-free environments. The obstacle avoidance is further considered in other papers, e.g., Lau et al. (2015), Restrepo et al. (2019), Roussos et al. (2008), Urakubo (2018), where a potential or navigation function is designed such that the robot is attracted to the desired configuration and repelled from obstacles. These results are more practical, but still have an inherent limitation that there may exist multiple critical points, i.e., the points where the gradient of the potential function vanishes. Moreover, these methods are difficult to apply to multi-robot systems, where the collision avoidance among robots is a critical requirement. Apart from stabilization, some papers such as Goodarzi et al. (2013), Lee et al. (2010), Yu and Ding (2016) study the simultaneous position and orientation tracking control, but the reference trajectory is assumed to be known in advance and planning is not considered.

To overcome the above-mentioned limitations, in this paper, we utilize the vector field (VF) method (Gao et al., 2023; Panagou, 2014; Zhang et al., 2009) to solve the motion planning problem for the 3D nonholonomic robots. Specifically, a velocity VF is defined over the workspace and the integral curve of the VF converges to the goal point. Compared to the optimization-based method, the VF motion planning specifies a reactive feedback in the sense that it is only related to the current state, while path replanning is not required. Furthermore, the VF directly specifies the heading direction at each point, which provides a reference to adjust the attitude of the robot so as to follow such a heading direction. In this way, the challenge caused by the nonholonomic constraints can be suitably handled via the VF method. Although several works have employed VF in the motion planning problem, such as He and Li (2024), Lindemann and LaValle (2009), Marchidan and Bakolas (2020), Panagou (2017), Yao et al. (2022), most of them focus on particle agents in 2D instead of rigid bodies in

3D. Additionally, the heading direction of the nonholonomic robot is rarely considered in the existing VF-based motion planning results.

Therefore, we investigate the motion planning problem of 3D nonholonomic robots via the VF method, where the terminal heading direction is specified. Moreover, the robot is modeled by a 6-DOF nonholonomic constrained rigid body rather than a particle agent. The contributions of this paper are summarized below.

1. Concerning the nonholonomic robot moving in an obstacle-free environment, we design a navigation VF (N-VF) which reaches any desired target position. More importantly, the N-VF points to a specified direction at the destination. Thus, by moving along the N-VF, the robot can reach the target position with the specified heading direction.
2. An obstacle avoidance VF (OA-VF) is proposed for motion planning in an obstacle-cluttered environment. The shape of the obstacle is described by the level surface of an implicit function, and the obstacle can be either static or moving. The OA-VF is composed of the N-VF and an additional VF tangential to the obstacle surface, and the composite OA-VF is free of any singularity.
3. For motion planning of multiple nonholonomic robots, a collision avoidance VF (CA-VF) is proposed to prevent the collisions among the robots. In order to reduce the motion coupling in collision avoidance, we propose a priority-based algorithm where the movement of higher prioritized robots is independent of the lower ones, so as to achieve motion decoupling.
4. To apply the proposed VF to the 3D rigid body, we design two auxiliary VFs which are always orthogonal to the N/OA/CA-VF. Moreover, such two auxiliary VFs are also orthogonal to each other. Therefore, these three VFs construct the basis of a body-fixed frame in $\mathbb{R}^3$, which can be further utilized as a reference for the attitude control of the robots.

Compared to the existing articles, which typically consider high-level planning and low-level control separately, this paper proposes a feedback motion planning method which achieves planning and control jointly for 3D nonholonomic robots. On one hand, different from the open-loop path or trajectory planning approaches, the proposed method does not have to design a tracking controller after the planning, but directly provide feedback control inputs based on the real-time planning results, i.e., the VFs, which have the capability to handle the nonholonomic constraints and henceforth guarantee the feasibility of motion planning. On the other hand, in contrast to the control-based methods such as those based on potential functions, the VFs presented in this paper give motion primitives of the 3D nonholonomic robot from any initial position, that is, the integral curves of the VFs. This is different from the pure feedback control approaches, which are usually hard to predict the path of the robot. Moreover, by designing the VFs straightforwardly, it is possible to guarantee that no extra critical points (except for the destination) exist in the state space of the robot, and the VFs proposed in this paper are singularity-free.

This paper is organized as follows. Section 2 provides the preliminaries and formulates the problems. The motion planning algorithms in obstacle-free and obstacle-cluttered environments are proposed in Section 3 and Section 4, respectively. Section 5 provides the cooperative motion planning algorithm for multiple nonholonomic robots. Numerical simulation results are given in Section 6, followed by Section 7, which concludes the paper.

Notations. Let $\dot{f}$ denote the time derivative $\frac{df(t)}{dt}$ of any differentiable function $f(t)$. The distance between a point $\bar{p} \in \mathbb{R}^n$ and

a nonempty set $S \subseteq \mathbb{R}^n$ is defined by $\text{dist}(\bar{\mathbf{p}}, S) := \inf\{\|\mathbf{p} - \bar{\mathbf{p}}\| : \mathbf{p} \in S\}$, where $\|\cdot\|$ denotes the Euclidean norm. The distance between two nonempty sets $A \subseteq \mathbb{R}^n$ and $B \subseteq \mathbb{R}^n$ is defined by $\text{dist}(A, B) := \inf\{\|\mathbf{a} - \mathbf{b}\| : \mathbf{a} \in A, \mathbf{b} \in B\}$. The notation $\mathbf{I}$ represents an identity matrix of suitable dimensions. The notations $\det$, tr and $\ker$ represent the determinant, trace and kernel of a matrix, respectively. The words “with respect to” is abbreviated as “w.r.t”.

2. Preliminaries and problem formulation

2.1. Nonholonomic robots modeling

Consider a swarm of nonholonomic robots labeled by $\mathcal{I}_V = \{1, \dots, N\}$, and each robot is modeled by a 6-DOF rigid body moving in the 3D Euclidean space $\mathbb{R}^3$. Let $\mathcal{F}_e$ denote the earth-fixed frame and $\mathcal{F}_{bi}$ denote the body-fixed frame of the i th nonholonomic robot ($i \in \mathcal{I}_V$), which is attached to the center of mass of the i th robot. Assume that each nonholonomic robot knows its position in $\mathcal{F}_e$ and its orientation relative to $\mathcal{F}_e$. The position of the i th nonholonomic robot in $\mathcal{F}_e$ is described by a vector $\mathbf{p}_i = [x_i \ y_i \ z_i]^T \in \mathbb{R}^3$, while the attitude is specified by a rotation matrix $\mathbf{R}_i \in \text{SO}(3) := \{\mathbf{R} \in \mathbb{R}^{3 \times 3} : \mathbf{R}^T \mathbf{R} = \mathbf{I}, \det(\mathbf{R}) = 1\}$, which depicts the rotation of $\mathcal{F}_{bi}$ relative to $\mathcal{F}_e$. The i th nonholonomic robot's angular velocity and linear velocity are denoted by $\boldsymbol{\Omega}_i = [\Omega_{xi} \ \Omega_{yi} \ \Omega_{zi}]^T \in \mathbb{R}^3$ and $\mathbf{v}_i = [v_{xi} \ v_{yi} \ v_{zi}]^T \in \mathbb{R}^3$, respectively, which are both provided in the body-fixed frame $\mathcal{F}_{bi}$. For all $\mathbf{a} = [a_x \ a_y \ a_z]^T \in \mathbb{R}^3$, we define the linear map “ $\wedge$ ” as follows

$$\mathbf{a}^\wedge = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \in \mathfrak{so}(3), \quad (1)$$

where $\mathfrak{so}(3) := \{\mathbf{S} \in \mathbb{R}^{3 \times 3} : \mathbf{S}^T = -\mathbf{S}\}$ denotes the Lie algebra of $\text{SO}(3)$. Herein, Lie algebra is a vector space endowed with a bilinear operation called *bracket* satisfying the anti-commutativity property and the Jacobi identity (Bullo & Lewis, 2005). Then, based on the definition of $\mathfrak{so}(3)$, we have the following property: For all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, it holds $\mathbf{a}^\wedge \mathbf{b} = \mathbf{a} \times \mathbf{b}$, where “ $\times$ ” denotes the cross product. Thus, the kinematics of the i th nonholonomic robot is given by

$$\dot{\mathbf{R}}_i = \mathbf{R}_i \boldsymbol{\Omega}_i^\wedge, \quad (2a)$$

$$\dot{\mathbf{p}}_i = \mathbf{R}_i \mathbf{v}_i, \quad (2b)$$

which describes the rotation and translation kinematics, respectively. Note that the motion of the nonholonomic robot is restricted by the nonholonomic constraints, so that the linear velocities along the y -axis and z -axis of the body-fixed frame $\mathcal{F}_{bi}$ are both zero, that is, $v_{yi} = v_{zi} = 0$. Hence, the nonholonomic robot with 6 DOFs is controlled by only 4 inputs, i.e., $\Omega_{xi}, \Omega_{yi}, \Omega_{zi}, v_{xi}$, demonstrating that the nonholonomic robot is a typical under-actuated system. As shown in (2), we consider the kinematics of the nonholonomic robot rather than the dynamics. This is because the nonholonomic constraint is a constraint on velocity (i.e., kinematic constraint) instead of acceleration, and the velocity in the kinematic model serves as the control input. Thus, it is sufficient to control a nonholonomic robot based on kinematics when we focus on the kinematic constraint.

Each nonholonomic robot has a *no-entering area*, which is a region that any other nonholonomic robot is strictly prohibited to enter; otherwise the collision would occur. Let $\mathcal{C}_{oi}$ denote the no-entering area of the i th nonholonomic robot and assume that $\mathcal{C}_{oi}$ is a sphere given by

$$\mathcal{C}_{oi} = \{\mathbf{p} \in \mathbb{R}^3 : \Psi_i(\mathbf{p}; \mathbf{p}_i) = \|\mathbf{p} - \mathbf{p}_i\| < r_c\}, \quad (3)$$

where $\mathbf{p}_i$ is the position of the i th nonholonomic robot and r_c is the radius of the sphere. Similarly, we introduce another area named *reactive area*, which indicates that a nonholonomic robot should take actions to avoid collisions once it enters the reactive area of other nonholonomic robots. Let $\mathcal{C}_{ri}$ denote the reactive area of the i th nonholonomic robot, which is defined by

$$\mathcal{C}_{ri} = \{\mathbf{p} \in \mathbb{R}^3 : r_c \leq \Psi_i(\mathbf{p}; \mathbf{p}_i) \leq r_d\}, \quad (4)$$

where r_d is the detection range of the nonholonomic robot, decided by the sensors of the robot. Note that the areas $\mathcal{C}_{oi}$ and $\mathcal{C}_{ri}$ are centered at $\mathbf{p}_i$, which is the position of the i th nonholonomic robot and possibly varies with time. Thus, $\mathcal{C}_{oi}$ and $\mathcal{C}_{ri}$ are both time-varying sets. But for the sake of simplicity, we omit the time parameter t in the paper.

2.2. Obstacles description

Consider a finite set of obstacles in the environment, and each of them is parameterized by a continuous function $\mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi}) : \mathbb{R}^3 \rightarrow \mathbb{R}$, where $\mathbf{p} = [x \ y \ z]^T \in \mathbb{R}^3$ is the position variable in $\mathcal{T}_i$ and $\mathbf{p}_{oi} = [x_{oi} \ y_{oi} \ z_{oi}]^T \in \mathbb{R}^3$ is either a constant or a time-varying vector denoting the center of the i th obstacle, $i \in \mathcal{I}_O = \{1, \dots, M\}$, and M is the number of the obstacles. Assume that $\mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi})$ has continuous first-order derivatives w.r.t the variable $\mathbf{p}$. In other words, the gradient $\nabla_{\mathbf{p}} \mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi})$ is continuous. Moreover, assume that there exists a positive constant $\bar{c}_i$ such that the level surface described by $\mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi}) = \bar{c}_i$ can enclose a region. Precisely, there exists a positive constant $\bar{c}_i$ such that the norm $\|\mathbf{p} - \mathbf{p}_{oi}\|$ is bounded for all $\mathbf{p}$ satisfying $\mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi}) = \bar{c}_i$. Without loss of generality, we utilize the equation $\mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi}) = 1$ to describe the surface of the obstacle, whose center is located at $\mathbf{p}_{oi}$. For instance, the surface $\mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi}) = \frac{(x-x_{oi})^2}{a^2} + \frac{(y-y_{oi})^2}{b^2} + \frac{(z-z_{oi})^2}{c^2} = 1$ represents an ellipsoid centered at (x_{oi}, y_{oi}, z_{oi}) with principal semi-axes of lengths a, b, c . The obstacle can be static or moving, which depends on the motion of $\mathbf{p}_{oi}$, and it is assumed that the obstacle's velocity $\dot{\mathbf{p}}_{oi}$ is known.

With the help of $\mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi})$, we can define the obstacle area $\mathcal{A}_{oi}$ and the reactive area $\mathcal{A}_{ri}$ as given below.

$$\mathcal{A}_{oi} = \{\mathbf{p} \in \mathbb{R}^3 : \mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi}) < 1\}, \quad (5)$$

$$\mathcal{A}_{ri} = \{\mathbf{p} \in \mathbb{R}^3 : 1 \leq \mathcal{T}_i(\mathbf{p}; \mathbf{p}_{oi}) \leq \bar{c}_i\}. \quad (6)$$

Since the obstacle's center $\mathbf{p}_o$ might be moving, the areas $\mathcal{A}_{oi}$ and $\mathcal{A}_{ri}$ could be time-varying sets. For convenience, the time parameter t is omitted. The obstacle area $\mathcal{A}_{oi}$ is the region where the obstacle occupies, and collision occurs if the nonholonomic robot enters $\mathcal{A}_{oi}$. Once in the reactive area $\mathcal{A}_{ri}$, the nonholonomic robot can sense the obstacle and needs to be reactive to the obstacle such that a potential collision can be avoided. Note that the size of the reactive area $\mathcal{A}_{ri}$ can be controlled by the constant $\bar{c}_i$. The following standing assumptions are imposed.

Assumption 1. It holds that $\mathbf{p}_{dj} \notin \bigcup_{i \in \mathcal{I}_O} (\mathcal{A}_{oi} \cup \mathcal{A}_{ri})$ for all time instant $t > 0$ and all $j \in \mathcal{I}_V$, where $\mathbf{p}_{dj} \in \mathbb{R}^3$ denotes the target position of the j th nonholonomic robot.

Assumption 2. It holds that $\text{dist}(\mathcal{A}_{ri}, \mathcal{A}_{rj}) > 0$ for all $i \neq j \in \mathcal{I}_O$ and $t > 0$.

Assumption 3. The reactive areas $\mathcal{C}_{ri}$ and $\mathcal{A}_{rj}$ are compact for all $i \in \mathcal{I}_V, j \in \mathcal{I}_O$ and $t > 0$.

Assumption 1 means that the target position of each robot cannot be covered by obstacles. Assumption 2 implies that any two obstacles are sufficiently far away such that their reactive areas do not overlap. Of course, if two obstacles in real-world

scenarios are too close such that [Assumption 2](#) is violated, then these two obstacles can be regarded as one big obstacle to make this assumption hold. [Assumption 3](#) stipulates that robots and obstacles are of finite sizes (i.e., bounded).

2.3. Problem formulation

Motion Planning Problem: Consider a swarm of nonholonomic robots labeled by $i \in \mathcal{I}_V$, and the kinematics of each robot is described by (2). Let $\mathbf{p}_{di} \in \mathbb{R}^3$ denote a target position and let the unit vector $\mathbf{e}_{di} \in \mathbb{R}^3$ denote a desired heading direction of the i th nonholonomic robot. Then, we design the angular velocity $\Omega_i^\wedge$ and the linear velocity $\mathbf{v}_i$ in (2), such that

1. $\lim_{t \rightarrow \infty} \|\mathbf{p}_i - \mathbf{p}_{di}\| = 0$ and $\lim_{t \rightarrow \infty} \|\mathbf{R}_i \mathbf{e}_x - \mathbf{e}_{di}\| = 0$ for all $i \in \mathcal{I}_V$, where $\mathbf{e}_x = [1 \ 0 \ 0]^T$;
2. $\text{dist}(\mathcal{C}_{oi}, \mathcal{A}_{oj}) > 0$ for all $i \in \mathcal{I}_V, j \in \mathcal{I}_O$ and $t > 0$;
3. $\text{dist}(\mathcal{C}_{oi}, \mathcal{C}_{oj}) > 0$ for all $i \neq j \in \mathcal{I}_V$ and $t > 0$.

In Objective 1, the first limit implies that the nonholonomic robot is driven to the target position $\mathbf{p}_{di}$. Regarding the second limit, due to the nonholonomic constraints, the robot's velocity direction is its heading direction, and the velocity in the body-fixed frame $\mathcal{F}_b$ is given by $\mathbf{v}_i = [v_{xi} \ 0 \ 0]^T$. Note that the direction of $\mathbf{v}_i$ can be represented by the unit vector $\mathbf{e}_x = [1 \ 0 \ 0]^T$, so that $\|\mathbf{R}_i \mathbf{e}_x - \mathbf{e}_{di}\|$ describes the error of directions between $\mathbf{v}_i$ and $\mathbf{e}_{di}$. Thus, the second limit in Objective 1 indicates that the velocity direction of the nonholonomic robot points to the desired heading direction $\mathbf{e}_{di}$ eventually. Objective 2 guarantees that the nonholonomic robot does not collide with the obstacles in the environment, and Objective 3 ensures that there does not exist any collisions among robots in the swarm.

We provide a definition which will be used later.

Definition 1. A VF $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ lies in a nonempty subspace $\Sigma \subseteq \mathbb{R}^3$ if it holds that $\mathbf{F}(\mathbf{p}) \in \Sigma$ for each point $\mathbf{p} \in \Sigma$. Namely, $\mathbf{F}(\Sigma) \subseteq \Sigma$.

It is obvious from [Definition 1](#) that if a VF $\mathbf{F}$ lies in a nonempty subspace Σ , then every complete trajectory of $\dot{\mathbf{p}}(t) = \mathbf{F}(\mathbf{p}(t))$ with the initial condition in Σ will stay in Σ for $t \geq 0$. In other words, if a VF $\mathbf{F}$ lies in Σ , then its integral curves are all contained in Σ .

3. Motion planning in obstacle-free environments

This section considers the motion planning of a single nonholonomic robot in an obstacle-free environment. We firstly design the VF which guides the motion of the nonholonomic robot to the target position and desired heading direction, and then derive the controller based on the proposed VF. Since only a single robot is considered in this section, for the sake of simplicity, we remove the subscript denoting the robot's label.

3.1. Vector field design

Let $\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$ denote the basis of the earth-fixed frame $\mathcal{F}_e$. We propose the navigation vector field (N-VF) $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $\mathbf{F} = F_x \mathbf{e}_x + F_y \mathbf{e}_y + F_z \mathbf{e}_z$, where the components are

$$F_x = x^2 - y^2 - z^2, \quad (7a)$$

$$F_y = 2xy, \quad (7b)$$

$$F_z = 2xz. \quad (7c)$$

Then, the N-VF $\mathbf{F}$ can guide the nonholonomic robot to the target position with the specified heading direction. Before proving this statement, we provide a property of $\mathbf{F}$ based on [Definition 1](#).

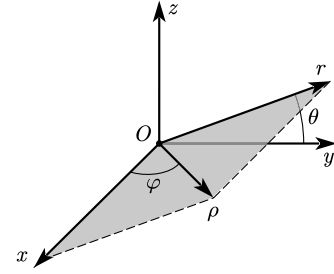


Fig. 1. Coordinate frames $\mathcal{F}_{xyz}, \mathcal{F}_{xr\theta}, \mathcal{F}_{\rho\varphi\theta}$.

Lemma 1. Given $a, b \in \mathbb{R}$, where $a^2 + b^2 \neq 0$, the N-VF $\mathbf{F}$ defined in (7) lies in the plane Σ_{ab} defined by $\Sigma_{ab} := \{(x, y, z) \in \mathbb{R}^3 : ay + bz = 0\}$. Therefore, Σ_{ab} is positively invariant w.r.t the dynamics $\dot{\mathbf{p}} = \mathbf{F}(\mathbf{p})$. Furthermore, the x -axis is positively invariant.

Proof. Based on (7), for all $\mathbf{p} \in \Sigma_{ab}$, it holds that

$$aF_y + bF_z = 2axy + 2bxz = 2x(ay + bz) = 0, \quad (8)$$

where $ay + bz = 0$ is utilized. It is implied by (8) that $\mathbf{F}(\mathbf{p}) \in \Sigma_{ab}$ for all $\mathbf{p} \in \Sigma_{ab}$. According to [Definition 1](#), we have that $\mathbf{F}$ lies in Σ_{ab} , and hence Σ_{ab} is positively invariant w.r.t $\dot{\mathbf{p}} = \mathbf{F}(\mathbf{p})$. In the x -axis, the components of $\mathbf{F}$ degenerate to $F_x = x^2, F_y = 0, F_z = 0$, which indicates that the x -axis is also positively invariant. ■

[Lemma 1](#) demonstrates that the N-VF $\mathbf{F}$ has a particular geometric structure, that is, it lies in a plane which passes through the x -axis. Motivated by this fact, we further investigate the characteristics of $\mathbf{F}$ in such a plane, which will be used in the proof of the main result. To this end, we reformulate the N-VF $\mathbf{F}$ under other coordinates in the 3D earth-fixed frame $\mathcal{F}_e$. Actually, besides the orthogonal coordinates (x, y, z) , $\mathcal{F}_e$ can also be described under the cylindrical coordinates or spherical coordinates. In the following, we let $\mathcal{F}_{\alpha\beta\gamma}$ represent the earth-fixed frame given by the coordinates (α, β, γ) . Then, $\mathcal{F}_{xyz}$ denotes the 3D orthogonal coordinate frame.

Define a new coordinate frame $\mathcal{F}_{xr\theta}$, and the coordinate transformation between $\mathcal{F}_{xyz}$ and $\mathcal{F}_{xr\theta}$ is defined by

$$x = x, \ y = r \cos \theta, \ z = r \sin \theta. \quad (9)$$

Based on $\mathcal{F}_{xr\theta}$, we define another coordinate frame $\mathcal{F}_{\rho\varphi\theta}$, and the transformation between $\mathcal{F}_{xr\theta}$ and $\mathcal{F}_{\rho\varphi\theta}$ is

$$x = \rho \cos \varphi, \ r = \rho \sin \varphi, \ \theta = \theta. \quad (10)$$

Remark 1. The coordinate frames $\mathcal{F}_{xyz}, \mathcal{F}_{xr\theta}, \mathcal{F}_{\rho\varphi\theta}$ are all earth-fixed frames, and the relations between these three frames are intuitively illustrated in [Fig. 1](#). The coordinate frame $\mathcal{F}_{xr\theta}$ actually defines a plane which passes through the x -axis and forms an angle of θ w.r.t the y -axis. Based on $\mathcal{F}_{xr\theta}$, the coordinate frame $\mathcal{F}_{\rho\varphi\theta}$ further provides the polar coordinates (ρ, φ) in the xOx plane.

Lemma 2. The N-VF $\mathbf{F}$ can be reformulated in $\mathcal{F}_{xr\theta}$ and $\mathcal{F}_{\rho\varphi\theta}$ as follows.

- (i) Let $\mathbf{e}_x, \mathbf{e}_r, \mathbf{e}_\theta$ denote the basis of $\mathcal{F}_{xr\theta}$, then $\mathbf{F}$ is expressed in the coordinate frame $\mathcal{F}_{xr\theta}$ as $\mathbf{F} = F_x \mathbf{e}_x + F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta$, where the components are

$$F_x = x^2 - r^2, \ F_r = 2xr, \ F_\theta = 0. \quad (11)$$

- (ii) Let $\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{e}_\theta$ denote the basis of $\mathcal{F}_{\rho\varphi\theta}$, then $\mathbf{F}$ is expressed in $\mathcal{F}_{\rho\varphi\theta}$ as $\mathbf{F} = F_\rho \mathbf{e}_\rho + F_\varphi \mathbf{e}_\varphi + F_\theta \mathbf{e}_\theta$, where the components are

$$F_\rho = \rho^2 \cos \varphi, \ F_\varphi = \rho^2 \sin \varphi, \ F_\theta = 0. \quad (12)$$

Proof. Please refer to [Appendix A](#). ■

Theorem 1. Let $\mathcal{X}_+$, $\mathcal{X}_-$ and $\mathcal{O}$ denote the positive x -semiaxis, the negative x -semiaxis and the origin of $\mathcal{F}_{xyz}$, respectively. The N-VF $\mathbf{F}$ defined in (7) has the following properties.

- (i) There is only one critical point of $\mathbf{F}$, that is, the origin $\mathcal{O}$, which is almost globally attractive w.r.t the dynamics $\dot{\mathbf{p}} = \mathbf{F}(\mathbf{p})$.
- (ii) Let ξ denote the integral curve of $\mathbf{F}$ starting from the initial conditions in $\mathbb{R}^3 \setminus \{\mathcal{X}_+ \cup \mathcal{O}\}$. Then, the tangent vector of ξ at the origin $\mathcal{O}$ points to the positive x -axis direction.

Proof. (i) It is evidently observed from (7) that $\mathbf{F}$ vanishes only at $\mathcal{O}$. Hence, $\mathcal{O}$ is the only critical point of $\mathbf{F}$, i.e., the only equilibrium of the dynamics $\dot{\mathbf{p}} = \mathbf{F}(\mathbf{p})$. To prove $\mathcal{O}$ is almost global attractive, we first compute the integral curves of $\mathbf{F}$. According to [Lemma 1](#), $\mathbf{F}$ lies in the plane Σ_{ab} , which can be expressed in $\mathcal{F}_{\rho\varphi\theta}$ as $\Sigma_{ab} : \theta - \theta_0 = 0$, where θ_0 is defined by $\theta_0 = \text{atan2}(-a, b)$. Then, in Σ_{ab} , the integral curve of $\mathbf{F}$ is the solution to the following ordinary differential equations

$$\frac{d\rho}{dt} = F_\rho, \quad \rho \frac{d\varphi}{dt} = F_\varphi. \quad (13)$$

If $\rho = 0$ (corresponding to $\mathcal{O}$), then $F_\rho = F_\varphi = F_\theta = 0$ and hence the origin is an equilibrium point. Now suppose $\rho \neq 0$ for the following discussion.

If $\varphi = 0$ (corresponding to any point on $\mathcal{X}_+$), then $F_\rho = \rho^2$, $F_\varphi = F_\theta = 0$. In this case, $\rho = x$. Therefore, by (13), one can directly obtain the analytic expression of any trajectory with the initial condition $(x_0, 0, 0)$, where $x_0 := x(0) > 0$, as below

$$x(t) = \frac{-1}{t - \frac{1}{x_0}} \quad (14)$$

and $y(t) = z(t) = 0$. Note that there is a finite escape time at $t = 1/x_0$. Therefore, any trajectory starting from $\mathcal{X}_+$ will escape to infinity in finite time.

If $\varphi = \pi$ (corresponding to any point on $\mathcal{X}_-$), then $F_\rho = -\rho^2$, $F_\varphi = F_\theta = 0$, and $\rho = -x$. It turns out that the analytic expression of any trajectory with the initial condition $(x_0, 0, 0)$, where $x_0 := x(0) < 0$, is still (14) and $y(t) = z(t) = 0$. However, in this case, there is no finite escape time since $x_0 < 0$, and instead, the trajectory will converge asymptotically to the origin $\mathcal{O}$ as $t \rightarrow \infty$.

If $\varphi \neq 0$ and $\varphi \neq \pi$, then it can be derived that

$$\frac{d\rho}{\rho d\varphi} = \frac{F_\rho}{F_\varphi} = \frac{\cos \varphi}{\sin \varphi}, \quad (15)$$

where (12) is utilized, and it follows that $(1/\rho)d\rho = \cot \varphi d\varphi$. By integrating it, we can obtain as below

$$\rho = C \sin \varphi, \quad (16)$$

where $C \in \mathbb{R} \setminus \{0\}$ is a constant. It should be mentioned that (16) represents a surface rather than a curve in the coordinate frame $\mathcal{F}_{\rho\varphi\theta}$. Let Λ denote the surface given in (16), and by coordinate transformations (9) and (10), it can be expressed in the coordinate frame $\mathcal{F}_{xyz}$ as

$$\Lambda : x^2 + y^2 + z^2 - C\sqrt{y^2 + z^2} = 0. \quad (17)$$

Since it has been proved that the N-VF $\mathbf{F}$ lies in the plane Σ_{ab} , then the integral curve of $\mathbf{F}$ is given by

$$\xi : \begin{cases} x^2 + y^2 + z^2 - C\sqrt{y^2 + z^2} = 0 \\ -\sin \theta_0 y + \cos \theta_0 z = 0 \end{cases}, \quad (18)$$

By substituting $(x, y, z) = (0, 0, 0)$ into it, we can verify that $\mathcal{O} \in \xi$. Regarding any initial condition $\mathbf{p}_0 \in \mathbb{R}^3 \setminus \{\mathcal{X}_+ \cup \mathcal{X}_- \cup \mathcal{O}\}$, it

is obvious that there exist a, b such that the plane Σ_{ab} contains $\mathbf{p}_0$ and Σ_{ab} is unique. Based on [Lemma 1](#), any trajectory starting from $\mathbf{p}_0$ will be contained in Σ_{ab} for $t \geq 0$, and the corresponding integral curve is described by (18), where θ_0 is determined by Σ_{ab} . Hence, the trajectory will asymptotically converge to $\mathcal{O}$ as $t \rightarrow \infty$. To sum up, any trajectory starting from $\mathbb{R}^3 \setminus (\mathcal{X}_+ \cup \mathcal{O})$ asymptotically converges to the origin $\mathcal{O}$. Namely, $\mathcal{O}$ is almost globally attractive as $\mathcal{X}_+$ is of measure zero.

(ii) We compute the tangent vector of ξ based on the formulation in (18). Since ξ is defined as the intersection of two surfaces, the tangent vector can be given by the cross product of the normal vectors of the two surfaces. Define $f_1 = x^2 + y^2 + z^2 - C\sqrt{y^2 + z^2}$ and $f_2 = -\sin \theta_0 y + \cos \theta_0 z$, and then the normal vectors of Λ and Σ_{ab} are given by

$$\begin{aligned} \mathbf{n}_\Lambda &= \frac{\partial f_1}{\partial x} \mathbf{e}_x + \frac{\partial f_1}{\partial y} \mathbf{e}_y + \frac{\partial f_1}{\partial z} \mathbf{e}_z \\ &= 2x\mathbf{e}_x + \left(2y - \frac{Cy}{\sqrt{y^2 + z^2}}\right) \mathbf{e}_y + \left(2z - \frac{Cz}{\sqrt{y^2 + z^2}}\right) \mathbf{e}_z, \\ \mathbf{n}_{\Sigma_{ab}} &= \frac{\partial f_2}{\partial x} \mathbf{e}_x + \frac{\partial f_2}{\partial y} \mathbf{e}_y + \frac{\partial f_2}{\partial z} \mathbf{e}_z \\ &= -\sin \theta_0 \mathbf{e}_y + \cos \theta_0 \mathbf{e}_z \\ &= -\frac{z}{\sqrt{y^2 + z^2}} \mathbf{e}_y + \frac{y}{\sqrt{y^2 + z^2}} \mathbf{e}_z. \end{aligned}$$

Having obtained $\mathbf{n}_\Lambda$ and $\mathbf{n}_{\Sigma_{ab}}$, we can calculate the tangent vector of the integral curve ξ as

$$\begin{aligned} \boldsymbol{\tau}_\xi &= \mathbf{n}_\Sigma \times \mathbf{n}_\Lambda \\ &= \left(C - 2\sqrt{y^2 + z^2}\right) \mathbf{e}_x + \frac{2xy}{\sqrt{y^2 + z^2}} \mathbf{e}_y + \frac{2xz}{\sqrt{y^2 + z^2}} \mathbf{e}_z \end{aligned}$$

One can calculate that

$$\lim_{x,y,z \rightarrow 0} \frac{2xy}{\sqrt{y^2 + z^2}} = 0, \quad \lim_{x,y,z \rightarrow 0} \frac{2xz}{\sqrt{y^2 + z^2}} = 0.$$

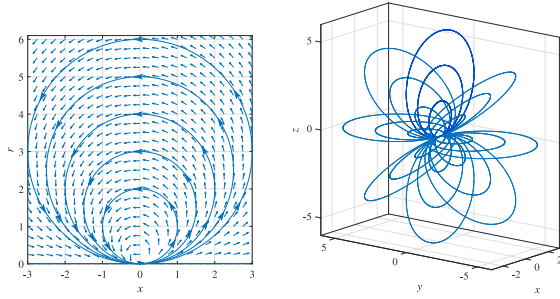
Therefore, the tangent vector $\boldsymbol{\tau}_\xi$ of the integral curve ξ at the origin of $\mathcal{F}_{xyz}$ is $\boldsymbol{\tau}_\xi|_{(x,y,z)=(0,0,0)} = C\mathbf{e}_x$, which indicates that the tangent vector $\boldsymbol{\tau}_\xi$ at the origin of $\mathcal{F}_{xyz}$ points along the x -axis. To further prove $\boldsymbol{\tau}_\xi$ points to the positive x -axis direction, we define $\Sigma_\epsilon = \{(x, y, z) \in \mathbb{R}^3 : |x| < \epsilon, y = z = 0\}$, where $\epsilon > 0$ is sufficiently small. Then, Σ_ϵ denotes a small neighborhood of the origin. Thus, it is obtained that $\mathbf{F}(\Sigma_\epsilon) = \epsilon^2 \mathbf{e}_x$, indicating the tangent vector points along the positive x -axis direction. ■

Remark 2. It is indicated by [Theorem 1](#) that the integral curve of the N-VF $\mathbf{F}$ always lies in the xOr plane, and the integral curve is expressed as (16) in the polar coordinates (ρ, φ) . By substituting (10) into (16), we can obtain the expression of the integral curve in orthogonal coordinates (x, r) , that is, $x^2 + (r - \frac{C}{2})^2 = (\frac{C}{2})^2$. This demonstrates that the integral curves in the xOr plane are circles located at $(x, r) = (0, \frac{C}{2})$ with radius $|\frac{C}{2}|$. [Fig. 2\(a\)](#) illustrates the N-VF $\mathbf{F}$ and its integral curve with $C > 0$ in the xOr plane. Once rotating the xOr plane around the x -axis, which is equivalent to choosing different values of θ , we can obtain the integral curves of $\mathbf{F}$ in the overall 3D space, as shown in [Fig. 2\(b\)](#).

[Theorem 1](#) can be extended to an arbitrary goal position and a goal orientation, which is given in the following corollary.

Corollary 1. Let $\mathbf{p}_d = [x_d \ y_d \ z_d]^T \in \mathbb{R}^3$ denote a desired position and $\mathbf{e}_d = \mathbf{R}_d \mathbf{e}_x$ a desired direction, where $\mathbf{R}_d \in \text{SO}(3)$ represents the rotation relative to the coordinate frame $\mathcal{F}_{xyz}$. Define the following N-VF

$$\tilde{\mathbf{F}} = \tilde{F}_x \mathbf{R}_d \mathbf{e}_x + \tilde{F}_y \mathbf{R}_d \mathbf{e}_y + \tilde{F}_z \mathbf{R}_d \mathbf{e}_z, \quad (19)$$



(a) Vector field in xOy plane. (b) Integral curves of vector field.

Fig. 2. Navigation vector field $\mathbf{F}$ and its integral curves.

where the components are

$$\tilde{F}_x = (x - x_d)^2 - (y - y_d)^2 - (z - z_d)^2, \quad (20a)$$

$$\tilde{F}_y = 2(x - x_d)(y - y_d), \quad (20b)$$

$$\tilde{F}_z = 2(x - x_d)(z - z_d). \quad (20c)$$

Then, almost every integral curve of the N-VF $\tilde{\mathbf{F}}$ passes through the position $\mathbf{p}_d$ and points along the direction of $\mathbf{e}_d$ simultaneously.

The N-VF $\tilde{\mathbf{F}}$ in (19) is obviously obtained by a coordinate transformation from the N-VF $\mathbf{F}$ in (7). Therefore, without loss of generality, we assume $\mathbf{p}_d = [0 \ 0 \ 0]^T$ and $\mathbf{e}_d = \mathbf{e}_x = [1 \ 0 \ 0]^T$ henceforth.

3.2. Controller design

Theorem 1 provides a VF which passes through a point with a desired direction. This naturally inspires us that we can let the nonholonomic robot follow the direction of the VF, so that it will arrive at the goal point with a specified heading direction. Note that the nonholonomic robot is a 3D rigid body, and its heading direction is the x -axis direction of the body-fixed frame $\mathcal{F}_b$. Although the VF in **Theorem 1** provides the x -axis direction of $\mathcal{F}_b$, the directions of y -axis and z -axis are undetermined yet, leading to the result that the angular velocity $\Omega^\wedge$ of the nonholonomic robot cannot be derived. Hence, in the following, we design two extra VFs which represent the y -axis and z -axis directions of the body-fixed frame $\mathcal{F}_b$, and then plus the N-VF $\mathbf{F}$ in **Theorem 1**, an auxiliary attitude matrix $\mathbf{R}_a$ can be constructed based on these three VFs. By tracking the attitude $\mathbf{R}_a$ with an angular velocity controller, the nonholonomic robot can align its heading direction with the N-VF $\mathbf{F}$, so as to accomplish motion planning.

Lemma 3. Define a VF $\mathbf{G}$ as follows

$$\mathbf{G} = G_x \mathbf{e}_x + G_y \mathbf{e}_y + G_z \mathbf{e}_z, \quad (21)$$

where the components are

$$G_x = 2x(y^2 + z^2), \quad (22a)$$

$$G_y = y(y^2 + z^2 - x^2), \quad (22b)$$

$$G_z = z(y^2 + z^2 - x^2). \quad (22c)$$

Then, $\mathbf{G}$ is in the xOy plane and orthogonal to the N-VF $\mathbf{F}$.

Proof. By substituting the coordinate transformation (9) and basis transformation (A.2) into (21) and (22), $\mathbf{G}$ can be rewritten in the frame $\mathcal{F}_{x\theta}$ as

$$\mathbf{G} = G_x \mathbf{e}_x + G_r \mathbf{e}_r + G_\theta \mathbf{e}_\theta, \quad (23)$$

where the components are

$$G_x = 2xr^2, \quad G_r = r(r^2 - x^2), \quad G_\theta = 0. \quad (24)$$

As illustrated in the proof of **Theorem 1**, $G_\theta = 0$ demonstrates that $\mathbf{G}$ lies in the xOy plane. To prove the orthogonality, we compute the dot product of $\mathbf{F}$ and $\mathbf{G}$, which is given by

$$\mathbf{F} \cdot \mathbf{G} = F_x G_x + F_r G_r + F_\theta G_\theta \quad (25)$$

Substituting (11) and (24) into (25), we have

$$\mathbf{F} \cdot \mathbf{G} = 0, \quad (26)$$

which implies the VFs $\mathbf{F}$ and $\mathbf{G}$ are orthogonal. ■

Based on $\mathbf{F}$ and $\mathbf{G}$, we define another VF $\mathbf{H}$ by the cross product of $\mathbf{F}$ and $\mathbf{G}$, that is

$$\mathbf{H} = \mathbf{G} \times \mathbf{F} = H_x \mathbf{e}_x + H_y \mathbf{e}_y + H_z \mathbf{e}_z, \quad (27)$$

where

$$H_x = 0, \quad (28a)$$

$$H_y = -z(x^2 + y^2 + z^2), \quad (28b)$$

$$H_z = y(x^2 + y^2 + z^2). \quad (28c)$$

Therefore, it follows from (26) and (27) that each two VFs of $\mathbf{F}, \mathbf{G}, \mathbf{H}$ are orthogonal, which indeed defines a 3D Cartesian coordinate frame at every point (x, y, z) in $\mathbb{R}^3$. Then, an attitude matrix $\mathbf{R}_a$ can be constructed to describe the attitude of such a coordinate frame. For simplicity, let vectors $\zeta_x, \zeta_y, \zeta_z$ denote the components of $\mathbf{F}, \mathbf{H}, \mathbf{G}$ at each point, and then it follows that

$$\zeta_x = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}, \quad \zeta_y = \begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix}, \quad \zeta_z = \begin{bmatrix} G_x \\ G_y \\ G_z \end{bmatrix}, \quad (29)$$

where the components are given in (7)(28)(22), respectively. Note that the vectors $\zeta_x, \zeta_y, \zeta_z$ are orthogonal to each other, based on which we can define the following auxiliary attitude matrix

$$\mathbf{R}_a = \begin{bmatrix} \frac{\zeta_x}{\|\zeta_x\|} & \frac{\zeta_y}{\|\zeta_y\|} & \frac{\zeta_z}{\|\zeta_z\|} \end{bmatrix} \in \text{SO}(3). \quad (30)$$

It can be observed from (7) that the N-VF $\mathbf{F}$ is singular, i.e., $F_x = F_y = F_z = 0$ only at the origin (the goal point). Thus, the auxiliary attitude matrix $\mathbf{R}_a$ is well defined almost everywhere except for the destination. However, by utilizing the trajectory's continuity, $\mathbf{R}_a$ can be suitably defined at the destination. The time derivative of $\mathbf{R}_a$ is

$$\dot{\mathbf{R}}_a = \begin{bmatrix} \frac{d}{dt} \left(\frac{\zeta_x}{\|\zeta_x\|} \right) & \frac{d}{dt} \left(\frac{\zeta_y}{\|\zeta_y\|} \right) & \frac{d}{dt} \left(\frac{\zeta_z}{\|\zeta_z\|} \right) \end{bmatrix}, \quad (31)$$

where $\frac{d}{dt} \left(\frac{\zeta_*}{\|\zeta_*\|} \right) = \frac{\dot{\zeta}_*}{\|\zeta_*\|} - (\zeta_*^T \dot{\zeta}_*) \frac{\zeta_*}{\|\zeta_*\|^3}$.

Lemma 4. Given the N-VF $\mathbf{F}$ in (7) and the auxiliary attitude $\mathbf{R}_a$ in (30), it holds $\|\mathbf{R}_a \mathbf{e}_x - \frac{\zeta_x}{\|\zeta_x\|}\| = 0$ if $\mathbf{R}_a^T \mathbf{R} = \mathbf{I}$, where $\mathbf{R}$ and $\mathbf{v}$ are the attitude and linear velocity of the nonholonomic robot, respectively.

Proof. Note that the equality $\mathbf{R}_a^T \mathbf{R} = \mathbf{I}$ implies $\mathbf{R} = \mathbf{R}_a$. Then, it follows that $\mathbf{R} \mathbf{e}_x = \mathbf{R}_a \mathbf{e}_x$, where $\mathbf{e}_x = [1 \ 0 \ 0]^T$. Considering the definition of $\mathbf{R}_a$ in (30), we have $\mathbf{R}_a \mathbf{e}_x = \frac{\zeta_x}{\|\zeta_x\|}$. Hence, it holds that

$$\mathbf{R} \mathbf{e}_x = \mathbf{R}_a \mathbf{e}_x = \frac{\zeta_x}{\|\zeta_x\|}, \text{ i.e., } \|\mathbf{R} \mathbf{e}_x - \frac{\zeta_x}{\|\zeta_x\|}\| = 0. \quad \blacksquare$$

Note that the vector ζ_x is indeed the direction of the N-VF $\mathbf{F}$. Then, **Lemma 4** demonstrates intuitively that the nonholonomic robot's heading direction is aligned with the N-VF $\mathbf{F}$ if the nonholonomic robot's attitude $\mathbf{R}$ tracks the auxiliary attitude $\mathbf{R}_a$. It is evident from **Theorem 1** that the nonholonomic robot will arrive at the goal point with the desired heading direction if it

moves along the N-VF $\mathbf{F}$, which is further converted by Lemma 4 to the attitude tracking of $\mathbf{R}$ w.r.t $\mathbf{R}_a$. In other words, once the nonholonomic robot tracks the attitude matrix $\mathbf{R}$, it can follow the N-VF $\mathbf{F}$ to reach the goal point and the desired orientation. Therefore, in the following, we will propose the control inputs of motion planning by solving the problem of attitude tracking.

Lemma 5. Given an auxiliary attitude $\mathbf{R}_a$, design the angular velocity $\Omega^\wedge$ as follows

$$\Omega^\wedge = -k_w \log_{\text{SO}(3)}(\mathbf{R}_a^{-1}\mathbf{R}) + \mathbf{R}^{-1}\dot{\mathbf{R}}_a\mathbf{R}_a^{-1}\mathbf{R}, \quad (32)$$

where $k_w > 0$ is the scalar control gain, and $\log_{\text{SO}(3)}$ is the logarithmic map on the Lie group $\text{SO}(3)$. Then, such a control law guarantees that the nonholonomic robot's attitude $\mathbf{R}$ exponentially tracks the auxiliary attitude $\mathbf{R}_a$ from all initial conditions $\mathbf{R}(0)$ satisfying $\text{tr}(\mathbf{R}_a^{-1}\mathbf{R}(0)) \neq -1$.

Proof. Define the attitude tracking error $\mathbf{R}_e = \mathbf{R}_a^{-1}\mathbf{R}$, and the time derivative of $\mathbf{R}_e$ is

$$\begin{aligned} \dot{\mathbf{R}}_e &= -\mathbf{R}_a^{-1}\dot{\mathbf{R}}_a\mathbf{R}_a^{-1}\mathbf{R} + \mathbf{R}_a^{-1}\dot{\mathbf{R}} \\ &= \mathbf{R}_e(\Omega^\wedge - \mathbf{R}^{-1}\dot{\mathbf{R}}_a\mathbf{R}_e), \end{aligned} \quad (33)$$

where the rotation kinematics of the nonholonomic robot is employed. Let $\Omega_e^\wedge$ denote the velocity of the error system, that is

$$\Omega_e^\wedge = \Omega^\wedge - \mathbf{R}^{-1}\dot{\mathbf{R}}_a\mathbf{R}_e. \quad (34)$$

Define $\Omega_a^\wedge = \mathbf{R}_a^T\dot{\mathbf{R}}_a$, and it is not difficult to verify that $\Omega_a^\wedge \in \mathfrak{so}(3)$. Then, it holds that $\dot{\mathbf{R}}_a = \mathbf{R}_a\Omega_a^\wedge$. By substituting it into (34), we have

$$\Omega_e^\wedge = \Omega^\wedge - \mathbf{R}^{-1}\mathbf{R}_a\Omega_a^\wedge\mathbf{R}_e = \Omega^\wedge - \text{Ad}_{\mathbf{R}_e^{-1}}\Omega_a^\wedge, \quad (35)$$

where the adjoint map $\text{Ad}_{\mathbf{R}} : \mathfrak{so}(3) \rightarrow \mathfrak{so}(3)$ is defined by $\text{Ad}_{\mathbf{R}}\Omega^\wedge = \mathbf{R}\Omega^\wedge\mathbf{R}^{-1}$, $\forall \Omega^\wedge \in \mathfrak{so}(3)$. Then, it is obtained that $\Omega_e^\wedge \in \mathfrak{so}(3)$. Thus, the error dynamics can be described by

$$\dot{\mathbf{R}}_e = \mathbf{R}_e\Omega_e^\wedge. \quad (36)$$

Based on the definition of $\mathbf{R}_e$, if it holds $\mathbf{R}_e = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix, then it is obtained that $\mathbf{R} = \mathbf{R}_a$, indicating that the mission of attitude tracking is achieved. It has been proved in Bullo and Murray (1995) that regarding the system $\dot{\mathbf{R}} = \mathbf{R}\Omega^\wedge$ in the Lie group $\text{SO}(3)$, the control law $\Omega^\wedge = -k_w \log_{\text{SO}(3)}(\mathbf{R})$ almost globally exponentially stabilizes the state $\mathbf{R}$ to $\mathbf{I}$, except for the initial conditions $\mathbf{R}_e(0)$ satisfying $\text{tr}(\mathbf{R}_e(0)) = -1$, where the logarithmic map $\log_{\text{SO}(3)}$ is undefined. Hence, for the purpose of attitude tracking, the error system's velocity $\Omega_e^\wedge$ can be designed as

$$\Omega_e^\wedge = -k_w \log_{\text{SO}(3)}(\mathbf{R}_e), \quad (37)$$

which is able to realize $\mathbf{R}_e \rightarrow \mathbf{I}$ exponentially. By substituting (37) into (34), we obtain the angular velocity controller of the nonholonomic robot as given in (32). ■

Regarding the control law of the linear velocity v_x , the distance error to the desired position (i.e., the origin of the earth-fixed frame) is introduced, so that v_x is given by

$$v_x = k_v \|\mathbf{p}\|. \quad (38)$$

To summarize, the control law for nonholonomic robot motion planning is presented in the following theorem.

Theorem 2. Regarding the nonholonomic robot (2) under the controller (32) and (38), it holds that $\lim_{t \rightarrow \infty} \|\mathbf{p}\| = 0$ and $\lim_{t \rightarrow \infty} \|\mathbf{R}_e - \mathbf{I}\| = 0$ for all initial position $\mathbf{p}(0) \in \mathbb{R}^3 \setminus \mathcal{X}_+$, where $\mathcal{X}_+ := \{(x, 0, 0) \in \mathbb{R}^3 : x > 0\}$, and all initial attitude $\mathbf{R}(0)$ satisfying $\text{tr}(\mathbf{R}_a^{-1}\mathbf{R}(0)) \neq -1$.

Proof. Based on Lemma 5, the angular velocity controller (32) guarantees that the attitude $\mathbf{R}$ exponentially tracks the auxiliary $\mathbf{R}_a$. By substituting (38) into (2b), the translational motion of the robot is given by $\dot{\mathbf{p}} = k_v \|\mathbf{p}\| \mathbf{R}_e \mathbf{e}_x$. According to Chicone (1999, Proposition 1.14), the integral curve of such a dynamical system is equivalent to that of $\dot{\mathbf{p}} = \mathbf{R}_e \mathbf{e}_x$, which is reformulated as

$$\dot{\mathbf{p}} = \mathbf{f}(t) + \mathbf{g}(t), \quad (39)$$

where $\mathbf{f}(t) := \mathbf{R}_a \mathbf{e}_x$ and $\mathbf{g}(t) := \mathbf{R}_e \mathbf{e}_x - \mathbf{R}_a \mathbf{e}_x$. Since $\mathbf{R}$ exponentially tracks $\mathbf{R}_a$, we obtain that $\mathbf{g}(t)$ vanishes exponentially. Let $\tilde{\xi}(t, \mathbf{p}(t_0), \mathbf{g}(t))$ denote the solution of the dynamics (39), where $\mathbf{p}(t_0)$ is the state at the initial time $t = t_0$. For $\mathbf{g}(t) \equiv \mathbf{0}$, we write $\xi(t, \mathbf{p}(t_0), \mathbf{g}(t))$ as $\xi(t, \mathbf{p}(t_0))$, which denotes the solution of the dynamics $\dot{\mathbf{p}} = \mathbf{f}(t)$. We recall that it has been given in Theorem 1 that $\xi(t, \mathbf{p}(t_0))$ asymptotically converges to the origin, and the region of attraction is given by $\mathcal{K} := \mathbb{R}^3 \setminus \mathcal{X}_+$.

We firstly prove that the trajectory $\tilde{\xi}(t, \mathbf{p}(0), \mathbf{g}(t))$ is bounded for $\mathbf{p}(0) \in \mathcal{K}$. It is estimated based on (39) that

$$\|\tilde{\xi}(t, \mathbf{p}(0), \mathbf{g}(t))\| \leq \|\xi(t, \mathbf{p}(0))\| + \int_0^t \|\mathbf{g}(\tau)\| d\tau. \quad (40)$$

Due to the exponential vanishing of $\mathbf{g}(t)$, there exist positive constants α and β such that $\|\mathbf{g}(t)\| \leq \alpha e^{-\beta t}$ for all $t \geq 0$. Note that α and β are independent of the initial position $\mathbf{p}(0)$, since they are determined by $\mathbf{g}(0)$, which represents the initial attitude error of $\mathbf{R}$ w.r.t $\mathbf{R}_a$. Thus, we obtain that $\int_0^t \|\mathbf{g}(\tau)\| d\tau \leq \int_0^t \alpha e^{-\beta \tau} d\tau = \frac{\alpha}{\beta} (1 - e^{-\beta t})$. Furthermore, it has been proved in Theorem 1 that for any $\mathbf{p}(0) \in \mathcal{K}$, the solution $\xi(t, \mathbf{p}(0))$ is bounded. Hence, for each $\mathbf{p}(0) \in \mathcal{K}$, there exist a $\delta_0 > 0$, such that $\|\xi(t, \mathbf{p}(0))\| \leq \delta_0$ for all $t \geq 0$. Thus, according to (40), we obtain that $\|\tilde{\xi}(t, \mathbf{p}(0), \mathbf{g}(t))\| \leq \delta_0 + \frac{\alpha}{\beta} (1 - e^{-\beta t})$ for all $t \geq 0$, which implies that $\tilde{\xi}(t, \mathbf{p}(0), \mathbf{g}(t))$ is bounded.

Next, we prove $\lim_{t \rightarrow \infty} \tilde{\xi}(t, \mathbf{p}(0), \mathbf{g}(t)) = \mathbf{0}$, which relies on the following lemma.

Lemma 6. For any $\epsilon \in (0, \frac{\alpha}{\beta})$, there exists a time instant $T > 0$, such that for any $\mathbf{p}(0) \in \mathcal{K}$, it holds that

$$\left| \|\tilde{\xi}(t, \mathbf{p}(0), \mathbf{g}(t))\| - \|\xi(t, \mathbf{p}(0))\| \right| \leq \epsilon, \quad \forall t \in [0, T]. \quad (41)$$

Proof of Lemma 6. Let ϵ be given. According to (39), it follows $\left| \|\tilde{\xi}(t, \mathbf{p}(0), \mathbf{g}(t))\| - \|\xi(t, \mathbf{p}(0))\| \right| \leq \int_0^t \|\mathbf{g}(\tau)\| d\tau \leq \frac{\alpha}{\beta} (1 - e^{-\beta t})$. Let $\frac{\alpha}{\beta} (1 - e^{-\beta t}) \leq \epsilon$, we obtain that $t \leq -\frac{1}{\beta} \ln(1 - \frac{\epsilon \beta}{\alpha})$. Define $T = -\frac{1}{\beta} \ln(1 - \frac{\epsilon \beta}{\alpha})$, and then the inequality (41) holds. Note that it should hold $\epsilon < \frac{\alpha}{\beta}$ such that $\ln(1 - \frac{\epsilon \beta}{\alpha})$ is well defined. Since α and β do not rely on $\mathbf{p}(0)$, then T only depends on ϵ . It has been proved that $\tilde{\xi}(t, \mathbf{p}(0), \mathbf{g}(t))$ and $\xi(t, \mathbf{p}(0))$ are bounded as long as $\mathbf{p}(0) \in \mathcal{K}$. Thus, $\mathbf{p}(0)$ can be arbitrarily chosen in $\mathcal{K}$. ■

Define $T_0 = 0$, and assume that a small $\epsilon_1 > 0$ and $\mathbf{p}(T_0) \in \mathcal{K}$ have been given. Based on Lemma 6, we have $\left| \|\tilde{\xi}(t, \mathbf{p}(T_0), \mathbf{g}(t))\| - \|\xi(t, \mathbf{p}(T_0))\| \right| \leq \epsilon_1$ for all $t \in [T_0, T_1]$, where $T_1 = -\frac{1}{\beta} \ln(1 - \frac{\epsilon_1 \beta}{\alpha})$. Then, we take T_1 as the initial time and $\mathbf{p}(T_1) := \tilde{\xi}(T_1, \mathbf{p}(T_0), \mathbf{g}(T_1))$ as the initial state. In order to employ Lemma 6 again, we have to ensure that $\mathbf{p}(T_1) \in \mathcal{K}$, or $\mathbf{p}(T_1) \notin \mathcal{X}_+$. Considering the case $\mathbf{p}(T_1) \in \mathcal{X}_+$, we have the following lemma.

Lemma 7. If $\mathbf{p}(T_1) \in \mathcal{X}_+$, the trajectory $\tilde{\xi}(t, \mathbf{p}(T_1), \mathbf{g}(t))$ will leave $\mathcal{X}_+$ at $T_1 + t_\epsilon$, where $t_\epsilon > 0$ is arbitrarily small.

Proof of Lemma 7. For $\mathbf{p}(T_1) \in \mathcal{X}_+$, the trajectory $\tilde{\xi}(t, \mathbf{p}(T_1), \mathbf{g}(t))$ will not leave $\mathcal{X}_+$ if and only if it holds that $\mathbf{g}_\perp(t) \equiv \mathbf{0}$ for all $t \geq T_1$, where $\mathbf{g}_\perp(t)$ denotes the component of $\mathbf{g}(t)$ perpendicular to

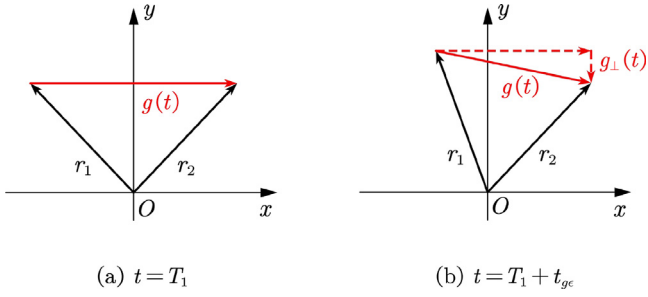


Fig. 3. $\mathbf{g}(t)$ at $t = T_1$ and $t = T_1 + t_{g\epsilon}$.

$\mathcal{X}_+$. If $\mathbf{g}_\perp(T_1) \neq 0$, Lemma 7 naturally holds, which is guaranteed by the continuity of $\xi(t, \mathbf{p}(T_1), \mathbf{g}(t))$ and $\mathbf{g}(t)$. Now, we consider the case $\mathbf{g}_\perp(T_1) = 0$. Note that $\mathbf{g}(t)$ is defined by $\mathbf{g}(t) = \mathbf{R}\mathbf{e}_x - \mathbf{R}_a\mathbf{e}_x$, where $\mathbf{R}\mathbf{e}_x$ and $\mathbf{R}_a\mathbf{e}_x$ are both unit vectors. Let $\mathbf{r}_1$ and $\mathbf{r}_2$ denote $\mathbf{R}\mathbf{e}_x$ and $\mathbf{R}_a\mathbf{e}_x$, respectively. Hence, $\mathbf{g}_\perp(T_1) = 0$ if and only if $\mathbf{r}_1 = [a_1 \ a_2 \ a_3]^T$ and $\mathbf{r}_2 = [\pm a_1 \ a_2 \ a_3]^T$, where a_i ($i = 1, 2, 3$) are constants satisfying $a_1^2 + a_2^2 + a_3^2 = 1$. If $\mathbf{r}_2 = [a_1 \ a_2 \ a_3]^T$, then $\mathbf{g}(T_1) = 0$, but this only occurs when $T_1 \rightarrow \infty$, and thus we can neglect this case and only focus on the case where $\mathbf{r}_2 = [-a_1 \ a_2 \ a_3]^T$. In this case, we have $\mathbf{g}(T_1) = [2a_1 \ 0 \ 0]^T$, implying that $\mathbf{g}(T_1)$ is parallel to $\mathcal{X}_+$, i.e., $\mathbf{g}_\perp(T_1) = 0$. Since $\mathbf{r}_1$ will converge to $\mathbf{r}_2$ and they are both unit vectors, it is evident that $\mathbf{r}_1$ begins to rotate to $\mathbf{r}_2$ at T_1 . This indicates that $\mathbf{g}(t)$ will have a non-zero component perpendicular to $\mathcal{X}_+$ at $T_1 + t_{g\epsilon}$, where $t_{g\epsilon} > 0$ is arbitrarily small, i.e., $\mathbf{g}_\perp(T_1 + t_{g\epsilon}) \neq 0$, as illustrated in Fig. 3. Therefore, $\xi(t, \mathbf{p}(T_1), \mathbf{g}(t))$ will leave $\mathcal{X}_+$ at a time instant larger than T_1 but infinitely close to T_1 . ■

According to Lemma 7, if $\mathbf{p}(T_1) \in \mathcal{X}_+$, it can be guaranteed that $\xi(t, \mathbf{p}(T_1), \mathbf{g}(t)) \notin \mathcal{X}_+$ at a time instant arbitrarily close to T_1 . Therefore, without loss of generality, we suppose that $\mathbf{p}(T_1) \in \mathcal{K}$. By taking $\mathbf{p}(T_1)$ as the initial state and using Lemma 6 again, we obtain that $|\|\xi(t, \mathbf{p}(T_1), \mathbf{g}(t))\| - \|\xi(t, \mathbf{p}(T_1))\|| \leq \epsilon_1$ for all $t \in [T_1, T_2]$, where $T_2 = \frac{1}{\beta} \ln(1 - \frac{2\epsilon_1\beta}{\alpha})$. Therefore, such an iteration demonstrates that $|\|\xi(t, \mathbf{p}(T_k), \mathbf{g}(t))\| - \|\xi(t, \mathbf{p}(T_k))\|| \leq \epsilon_1$, for all $t \in [T_k, T_{k+1}]$, where $T_k = -\frac{1}{\beta} \ln(1 - \frac{k\epsilon_1\beta}{\alpha})$, $k = 0, 1, 2, \dots$. Based on Theorem 1, the equilibrium of $\dot{\mathbf{p}} = \mathbf{f}(t)$ is almost global attractive. Then, for any $\epsilon_2 > 0$ and for any $\mathbf{p}(t_0) \in \mathcal{K}$, there exists a time instant $\bar{T} > t_0$, such that $\|\xi(t, \mathbf{p}(t_0))\| < \epsilon_2$ for all $t \geq \bar{T}$, where t_0 denotes the initial time. According to the definition of T_k , the interval $T_{k+1} - T_k$ becomes larger as k increases. Therefore, once we let $\epsilon_1 = \epsilon_2 = \epsilon$, where $\epsilon > 0$ is arbitrarily small, there exist a time instant $\bar{T} > 0$ and $\bar{k} \in \{0, 1, 2, \dots\}$ with the property $T_{\bar{k}} \leq \bar{T} < T_{\bar{k}+1}$, such that $\|\xi(t, \mathbf{p}(T_{\bar{k}}))\| < \epsilon$ for all $t \geq \bar{T}$. Moreover, for $k = \bar{k}$, it holds that $|\|\xi(t, \mathbf{p}(T_{\bar{k}}), \mathbf{g}(t))\| - \|\xi(t, \mathbf{p}(T_{\bar{k}}))\|| \leq \epsilon$ for all $t \in [T_{\bar{k}}, T_{\bar{k}+1}]$. Thus, $\xi(t, \mathbf{p}(T_{\bar{k}}), \mathbf{g}(t))$ satisfies

$$\|\xi(t, \mathbf{p}(T_{\bar{k}}), \mathbf{g}(t))\| \leq \|\xi(t, \mathbf{p}(T_{\bar{k}}))\| + \epsilon < 2\epsilon, \quad \forall t \in [\bar{T}, T_{\bar{k}+1}].$$

Now, start from the state $\mathbf{p}(T_{\bar{k}+1})$. Since $\|\xi(t, \mathbf{p}(T_{\bar{k}}))\| < \epsilon$ holds for all $t \geq \bar{T}$, it can be similarly obtained that $\|\xi(t, \mathbf{p}(T_{\bar{k}+1}), \mathbf{g}(t))\| < 2\epsilon$ for all $t \in [T_{\bar{k}+1}, T_{\bar{k}+2}]$. Then, an induction argument gives that $\|\xi(t, \mathbf{p}(T_{\bar{k}}), \mathbf{g}(t))\| < 2\epsilon$ for all $t \geq \bar{T}$, where ϵ is arbitrarily small. This proves that the position of the closed-loop system asymptotically converges to the origin, i.e., $\lim_{t \rightarrow \infty} \xi(t, \mathbf{p}(0), \mathbf{g}(t)) = 0$.

Finally, we prove $\lim_{t \rightarrow \infty} \|\mathbf{R}\mathbf{e}_x - \mathbf{e}_x\| = 0$. Due to the fact that $\mathbf{R}$ exponentially tracks $\mathbf{R}_a$, it is obtained that $\lim_{t \rightarrow \infty} \|\mathbf{R}\mathbf{e}_x - \mathbf{e}_x\| = \|\mathbf{R}_a\mathbf{e}_x - \mathbf{e}_x\|$, where $\mathbf{R}_a\mathbf{e}_x$ is the tangent vector of the trajectory ξ at the final position. Note that ξ converges to the origin eventually,

and Theorem 1 guarantees that $\mathbf{R}_a\mathbf{e}_x = \mathbf{e}_x$ at the origin. Hence, it follows that $\lim_{t \rightarrow \infty} \|\mathbf{R}\mathbf{e}_x - \mathbf{e}_x\| = \|\mathbf{R}_a\mathbf{e}_x - \mathbf{e}_x\| = 0$. ■

Remark 3. Theorem 2 provides the control law in the case where the goal point is the origin and goal direction is the x -axis. Regarding arbitrary desired point and direction, Corollary 1 has provided the related N-VF $\tilde{\mathbf{F}}$. Then, the angular velocity in (32) can be correspondingly revised by reconstructing the auxiliary attitude matrix $\mathbf{R}_a$ with the aid of $\tilde{\mathbf{F}}$ in (19). Besides, the linear velocity in (38) can also be reformulated to be $v_x = k_v \|\mathbf{p} - \mathbf{p}_d\|$.

Remark 4. The formulation in (22) and (28) is merely one possible way to construct two auxiliary VFs $\mathbf{G}$ and $\mathbf{H}$, which in fact represent the directions of z -, y -axis of the body-fixed frame $\mathcal{F}_b$. Since there only exists a requirement for the nonholonomic robot's heading direction (i.e., the direction of the x -axis), the directions of the y -, z -axis can be freely designed as long as they can form a set of basis in the 3D Cartesian frame. Of course, it should be mentioned that the VFs $\mathbf{G}$ and $\mathbf{H}$ presented in this paper have explicit geometric meanings. More specifically, $\mathbf{G}$ represents the outward normal vector of the integral curves in the xOy plane (the integral curves are the circles shown in Fig. 2(a)), while $\mathbf{H}$ represents the normal vector of the xOz plane. Note that there are other approaches to construct $\mathbf{G}$ and $\mathbf{H}$; on the premise of orthogonality, we could design $\mathbf{G}$ and $\mathbf{H}$ by further considering the full orientation constraints at the final time instant, so that all the attitude DOFs can be exactly planned.

4. Motion planning in obstacle-cluttered environments

In this section, we consider the motion planning of a non-holonomic robot in an obstacle-cluttered environment. According to Assumption 2, the nonholonomic robot can enter the reactive area of one obstacle at each time instant. Therefore, for the sake of simplicity, we remove the subscript of the obstacle's label and consider the problem of circumventing one obstacle. In addition, the subscript of the robot's label is also removed since only one robot is considered. Based on Section 2.3, the obstacle avoidance problem is mathematically formulated by $\text{dist}(\mathcal{C}_0, \mathcal{A}_0) > 0$, where $\mathcal{C}_0$ is the robot's no-entering area given in (3) and $\mathcal{A}_0$ is the obstacle area defined in (5). Note that the robot's no-entering area $\mathcal{C}_0$ is a sphere with radius r_c . By adjusting the parameter $\bar{c}$ of the level surface $\mathcal{Y}(\mathbf{p}; \mathbf{p}_0) = \bar{c}$, the obstacle area $\mathcal{A}_0$ can be enlarged radially with r_c . In this way, the objective of obstacle avoidance can be reformulated by $\text{dist}(\mathbf{p}, \mathcal{A}_0) > 0$, where $\mathbf{p}$ is the position of the nonholonomic robot. Therefore, in the following, the obstacle avoidance VF (OA-VF) is presented based on the requirement of $\text{dist}(\mathbf{p}, \mathcal{A}_0) > 0$.

To design the OA-VF, we firstly compute the normal vector at each point on the surface $\mathcal{Y}(\mathbf{p}; \mathbf{p}_0) = k$ ($1 \leq k \leq \bar{c}$), which is given by

$$\mathbf{n}_\mathcal{Y} = \frac{\partial \mathcal{Y}}{\partial x} \mathbf{e}_x + \frac{\partial \mathcal{Y}}{\partial y} \mathbf{e}_y + \frac{\partial \mathcal{Y}}{\partial z} \mathbf{e}_z. \quad (42)$$

The time derivative of $\mathbf{n}_\mathcal{Y}$ is

$$\dot{\mathbf{n}}_\mathcal{Y} = \frac{\partial \mathbf{n}_\mathcal{Y}}{\partial \mathbf{p}} \dot{\mathbf{p}} + \frac{\partial \mathbf{n}_\mathcal{Y}}{\partial \mathbf{p}_0} \dot{\mathbf{p}}_0. \quad (43)$$

Then, based on the normal vector $\mathbf{n}_\mathcal{Y}$, we propose the OA-VF according to two different cases, which are classified by whether the normal vector $\mathbf{n}_\mathcal{Y}$ is collinear with the N-VF $\mathbf{F}$ given in (7). In the following, we assume the target point $\mathbf{p}_d$ is the origin of the earth-fixed frame $\mathcal{F}_e$ for simplicity.

(1) $\mathbf{n}_\mathcal{Y}$ and $\mathbf{F}$ are not collinear ($\mathbf{n}_\mathcal{Y} \nparallel \mathbf{F}$)

Define a new vector $\tau_\mathcal{Y}^a$ by the following cross product

$$\tau_\mathcal{Y}^a = \mathbf{n}_\mathcal{Y} \times \mathbf{F}. \quad (44)$$

Substituting (7) and (42) into (44), we obtain that

$$\tau_{\mathcal{F}}^a = T_x^a \mathbf{e}_x + T_y^a \mathbf{e}_y + T_z^a \mathbf{e}_z, \quad (45)$$

where the components are given by $T_x^a = \frac{\partial \mathcal{F}}{\partial y} F_z - \frac{\partial \mathcal{F}}{\partial z} F_y$, $T_y^a = \frac{\partial \mathcal{F}}{\partial z} F_x - \frac{\partial \mathcal{F}}{\partial x} F_z$, $T_z^a = \frac{\partial \mathcal{F}}{\partial x} F_y - \frac{\partial \mathcal{F}}{\partial y} F_x$. Then, we define another vector $\tau_{\mathcal{F}}^b$ as follows

$$\tau_{\mathcal{F}}^b = \tau_{\mathcal{F}}^a \times \mathbf{n}_{\mathcal{F}}. \quad (46)$$

By substituting (42) and (44) into (46), it is obtained that

$$\tau_{\mathcal{F}}^b = T_x^b \mathbf{e}_x + T_y^b \mathbf{e}_y + T_z^b \mathbf{e}_z, \quad (47)$$

where the components are given by $T_x^b = ((\frac{\partial \mathcal{F}}{\partial z})^2 + (\frac{\partial \mathcal{F}}{\partial y})^2) F_x - \frac{\partial \mathcal{F}}{\partial x} (\frac{\partial \mathcal{F}}{\partial z} F_z + \frac{\partial \mathcal{F}}{\partial y} F_y)$, $T_y^b = ((\frac{\partial \mathcal{F}}{\partial x})^2 + (\frac{\partial \mathcal{F}}{\partial z})^2) F_y - \frac{\partial \mathcal{F}}{\partial y} (\frac{\partial \mathcal{F}}{\partial x} F_x + \frac{\partial \mathcal{F}}{\partial z} F_z)$, $T_z^b = ((\frac{\partial \mathcal{F}}{\partial y})^2 + (\frac{\partial \mathcal{F}}{\partial x})^2) F_z - \frac{\partial \mathcal{F}}{\partial z} (\frac{\partial \mathcal{F}}{\partial y} F_y + \frac{\partial \mathcal{F}}{\partial x} F_x)$.

(2) $\mathbf{n}_{\mathcal{F}}$ and $\mathbf{F}$ are collinear ($\mathbf{n}_{\mathcal{F}} \parallel \mathbf{F}$)

When $\mathbf{n}_{\mathcal{F}}$ and $\mathbf{F}$ are lying in the same straight line, the VF $\tau_{\mathcal{F}}^a$ is undefined since the cross product in (44) is $\mathbf{0}$. In this case, we directly define

$$\tau_{\mathcal{F}}^a = \mathbf{H}, \quad (48)$$

$$\tau_{\mathcal{F}}^b = \mathbf{G}, \quad (49)$$

where $\mathbf{G}$ and $\mathbf{H}$ are VFs given in (21) and (27).

From the geometrical point of view, $\mathbf{n}_{\mathcal{F}}$ is the normal vector of the obstacle surface, while $\tau_{\mathcal{F}}^a$ and $\tau_{\mathcal{F}}^b$ are the tangential vectors lying in the plane which is orthogonal to $\mathbf{n}_{\mathcal{F}}$. Moreover, $\tau_{\mathcal{F}}^a$ and $\tau_{\mathcal{F}}^b$ are also orthogonal to each other. Thus, based on the N-VF $\mathbf{F}$ and the tangential VF $\tau_{\mathcal{F}}^b$, we propose the OA-VF in the reactive area $\mathcal{A}_r$ as follows

$$\mathbf{F}_{OA} = \chi \mathbf{F} + (1 - \chi) \tau_{\mathcal{F}}^b, \quad (50)$$

where χ is a smooth function defined by

$$\chi = \begin{cases} 0, & \mathbf{p} \in \{\mathcal{T}(\mathbf{p}; \mathbf{p}_0) \leq 1\} \\ S(\mathbf{p}), & \mathbf{p} \in \{1 < \mathcal{T}(\mathbf{p}; \mathbf{p}_0) < \bar{c}\} \\ 1, & \mathbf{p} \in \{\mathcal{T}(\mathbf{p}; \mathbf{p}_0) \geq \bar{c}\} \end{cases} \quad (51)$$

and $S(\mathbf{p})$ is a smooth function valued in $(0, 1)$, which can be constructed by the typical bump functions (Fry & McManus, 2002).

The primary requirement for $\mathbf{F}_{OA}$ is to guarantee the obstacle avoidance. Besides, since $\mathbf{F}_{OA}$ is a composite VF of $\mathbf{F}$ and $\tau_{\mathcal{F}}^b$, we should first investigate whether $\mathbf{F}_{OA}$ has singular points where the nonholonomic robot would get stuck possibly.

Theorem 3. The OA-VF $\mathbf{F}_{OA}$ proposed in (50) has the following properties:

- (i) $\mathbf{F}_{OA}$ never penetrates the obstacle area $\mathcal{A}_0$;
- (ii) $\mathbf{F}_{OA}$ does not vanish in the reactive area $\mathcal{A}_r$.

Proof. (i) Impenetrability of the obstacle area $\mathcal{A}_0$ can be guaranteed if the OA-VF $\mathbf{F}_{OA}$ projects zero onto the normal vector $\mathbf{n}_{\mathcal{F}}$ of the obstacle surface $\mathcal{T}(\mathbf{p}; \mathbf{p}_0) = 1$. In light of this fact, we should compute the dot product of $\mathbf{F}_{OA}$ and $\mathbf{n}_{\mathcal{F}}$. According to (50), the OA-VF $\mathbf{F}_{OA}$ on the obstacle surface $\mathcal{T}(\mathbf{p}; \mathbf{p}_0) = 1$ is equivalent to $\mathbf{F}_{OA}|_{\mathcal{T}=1} = \tau_{\mathcal{F}}^b$. Then, it follows that

$$(\mathbf{F}_{OA} \cdot \mathbf{n}_{\mathcal{F}})|_{\mathcal{T}=1} = \tau_{\mathcal{F}}^b \cdot \mathbf{n}_{\mathcal{F}}. \quad (52)$$

For the case of $\mathbf{n}_{\mathcal{F}} \nparallel \mathbf{F}$, we substitute (46) into (52), and it is obtained that

$$(\mathbf{F}_{OA} \cdot \mathbf{n}_{\mathcal{F}})|_{\mathcal{T}=1} = (\tau_{\mathcal{F}}^a \times \mathbf{n}_{\mathcal{F}}) \cdot \mathbf{n}_{\mathcal{F}} = -(\mathbf{n}_{\mathcal{F}} \times \mathbf{n}_{\mathcal{F}}) \cdot \tau_{\mathcal{F}}^a = 0, \quad (53)$$

where the property of mixed product is utilized. For the case of $\mathbf{n}_{\mathcal{F}} \parallel \mathbf{F}$, we have

$$(\mathbf{F}_{OA} \cdot \mathbf{n}_{\mathcal{F}})|_{\mathcal{T}=1} = \tau_{\mathcal{F}}^b \cdot \mathbf{n}_{\mathcal{F}} = \mathbf{G} \cdot \mathbf{n}_{\mathcal{F}}, \quad (54)$$

where (49) is employed. Due to $\mathbf{n}_{\mathcal{F}} \parallel \mathbf{F}$, the normal vector $\mathbf{n}_{\mathcal{F}}$ can be expressed as $\mathbf{n}_{\mathcal{F}} = k\mathbf{F}$, where k is a nonzero scalar. By substitute it into (54), we have

$$(\mathbf{F}_{OA} \cdot \mathbf{n}_{\mathcal{F}})|_{\mathcal{T}=1} = k\mathbf{G} \cdot \mathbf{F} = 0, \quad (55)$$

which is guaranteed by (26). Eqs. (53) and (55) indicate that the OA-VF $\mathbf{F}_{OA}$ is orthogonal to the normal vector $\mathbf{n}_{\mathcal{F}}$ on the obstacle surface. In other words, the $\mathbf{F}_{OA}$ lies in the tangential plane of the obstacle surface, so that it will never penetrate the obstacle area $\mathcal{A}_0$.

(ii) We first consider the case of $\mathbf{n}_{\mathcal{F}} \nparallel \mathbf{F}$. For simplicity, we use the components to represent the VFs $\mathbf{n}_{\mathcal{F}}$ and $\mathbf{F}$ in the following. To be more specific, they are denoted by $\mathbf{n}_{\mathcal{F}} = [\frac{\partial \mathcal{F}}{\partial x} \frac{\partial \mathcal{F}}{\partial y} \frac{\partial \mathcal{F}}{\partial z}]^T$ and $\mathbf{F} = [F_x F_y F_z]^T$, respectively. Under such a formulation, the cross product “ $\times$ ” can be expressed as the hat map “ $\wedge$ ” given in (1). Then, $\tau_{\mathcal{F}}^a$ in (44) can be rewritten as $\tau_{\mathcal{F}}^a = \mathbf{n}_{\mathcal{F}}^\wedge \mathbf{F}$, and $\tau_{\mathcal{F}}^b$ in (46) can be rewritten as

$$\tau_{\mathcal{F}}^b = (\mathbf{n}_{\mathcal{F}}^\wedge \mathbf{F}) \times \mathbf{n}_{\mathcal{F}} = -\mathbf{n}_{\mathcal{F}}^\wedge (\mathbf{n}_{\mathcal{F}}^\wedge \mathbf{F}) = -(\mathbf{n}_{\mathcal{F}}^\wedge)^2 \mathbf{F}. \quad (56)$$

By substituting (56) into (50), $\mathbf{F}_{OA}$ can be reorganized as

$$\mathbf{F}_{OA} = \chi \mathbf{F} + (\chi - 1)(\mathbf{n}_{\mathcal{F}}^\wedge)^2 \mathbf{F} = \Xi_{OA} \mathbf{F}, \quad (57)$$

where Ξ_{OA} is a matrix defined by

$$\Xi_{OA} = \chi \mathbf{I} + (\chi - 1)(\mathbf{n}_{\mathcal{F}}^\wedge)^2, \quad (58)$$

and $\mathbf{I}$ is the identity matrix. Based on the formulations given in (7), $\mathbf{F}$ only vanishes as the desired point $\mathbf{p}_d$. According to Assumption 1, there does not exist $\mathbf{p}_d$ in the reactive area $\mathcal{A}_r$, implying that $\mathbf{F} \neq \mathbf{0}$ in $\mathcal{A}_r$. Therefore, referring to (57), $\mathbf{F}_{OA}$ does not vanish in $\mathcal{A}_r$ if and only if the matrix Ξ_{OA} is invertible. By calculation, the determinant of Ξ_{OA} is

$$\det(\Xi_{OA}) = \chi + 2\chi^2(1 - \chi)\|\mathbf{n}_{\mathcal{F}}\|^2 + \chi(1 - \chi)^2\|\mathbf{n}_{\mathcal{F}}\|^4. \quad (59)$$

Due to $\|\mathbf{n}_{\mathcal{F}}\| \neq 0$, it is obvious that $\det(\Xi_{OA}) \neq 0$ for $0 < \chi \leq 1$, indicating that the matrix Ξ_{OA} is invertible in $\mathcal{A}_r/\{\mathcal{T}(\mathbf{p}; \mathbf{p}_0) = 1\}$. Once $\mathbf{p} \in \{\mathcal{T}(\mathbf{p}; \mathbf{p}_0) = 1\}$, i.e., $\chi = 0$, there hold $\det(\Xi_{OA}) = 0$, implying the matrix Ξ_{OA} will lose rank. In this case, $\mathbf{F}_{OA}$ does not vanish if and only if $\mathbf{F}$ does not lie in $\ker(\Xi_{OA})$. When $\chi = 0$, Ξ_{OA} degenerates to

$$\Xi_{OA} = -(\mathbf{n}_{\mathcal{F}}^\wedge)^2. \quad (60)$$

Thus, it can be derived that

$$\ker(\Xi_{OA}) = \ker(\mathbf{n}_{\mathcal{F}}^\wedge) = \{\alpha \in \mathbb{R}^3 : \alpha \parallel \mathbf{n}_{\mathcal{F}}\}. \quad (61)$$

Note that the premise of this case is $\mathbf{n}_{\mathcal{F}} \nparallel \mathbf{F}$, so that $\mathbf{F} \notin \ker(\Xi_{OA})$ indicating that $\mathbf{F}_{OA} \neq \mathbf{0}$ on the obstacle surface $\mathcal{T}(\mathbf{p}; \mathbf{p}_0) = 1$. Therefore, $\mathbf{F}_{OA}$ does not vanish in the reactive area $\mathcal{A}_r$ for the case of $\mathbf{n}_{\mathcal{F}} \nparallel \mathbf{F}$.

Next, we deal with the case of $\mathbf{n}_{\mathcal{F}} \parallel \mathbf{F}$. In this case, owing to the formulation of $\tau_{\mathcal{F}}^b$ in (49), $\mathbf{F}_{OA}$ in (50) degenerates to

$$\mathbf{F}_{OA} = \chi \mathbf{F} + (1 - \chi) \mathbf{G}. \quad (62)$$

According to Lemma 3, the VFs $\mathbf{F}$ and $\mathbf{G}$ are linearly independent. Hence, as the linear combination of $\mathbf{F}$ and $\mathbf{G}$, the OA-VF $\mathbf{F}_{OA}$ does not vanish everywhere in the reactive area $\mathcal{A}_r$. ■

Remark 5. One can observe the following two properties:

- (i) $\mathbf{F}$, $\mathbf{n}_{\mathcal{F}}$ and $\tau_{\mathcal{F}}^b$ are always in the same plane since they are all orthogonal to $\tau_{\mathcal{F}}^a$;
- (ii) The angle between $\mathbf{F}$ and $\tau_{\mathcal{F}}^b$ is always less than or equal to $\pi/2$, due to the fact that $\mathbf{F}^T \tau_{\mathcal{F}}^b \stackrel{(56)}{=} -\mathbf{F}^T (\mathbf{n}_{\mathcal{F}}^\wedge)^2 \mathbf{F} = \mathbf{F}^T \mathbf{n}_{\mathcal{F}}^\wedge \mathbf{n}_{\mathcal{F}}^\wedge \mathbf{F} = \|\mathbf{n}_{\mathcal{F}}^\wedge \mathbf{F}\|^2 \geq 0$, where the equality holds only in the case of $\mathbf{n}_{\mathcal{F}} \parallel \mathbf{F}$.

Based on [Theorem 3](#), in order to avoid obstacles, the OA-VF $\mathbf{F}_{OA}$ can be employed as the nonholonomic robot's heading direction (i.e., the x-axis direction of the body-fixed frame) in the reactive area $\mathcal{A}_r$. However, as shown in the free-space motion planning, it is necessary to design another two vector fields to determine the y-axis and z-axis directions of the body-fixed frame $\mathcal{F}_e$, and the requirement for these three VFs is that each two of them should be orthogonal. The following lemma gives the formulation of the other two VFs.

Lemma 8. Define two VFs $\mathbf{G}_{OA}$ and $\mathbf{H}_{OA}$ by

$$\mathbf{G}_{OA} = \mu \mathbf{G} + (1 - \mu) \mathbf{n}_r, \quad (63)$$

$$\mathbf{H}_{OA} = \chi \mu \mathbf{H} + (1 - \mu) \boldsymbol{\tau}_r^a + \mu(1 - \chi) \mathbf{G} \times \boldsymbol{\tau}_r^b, \quad (64)$$

where μ is given by

$$\mu = \frac{\chi \mathbf{F} \cdot \mathbf{n}_r}{\chi \mathbf{F} \cdot \mathbf{n}_r + (\chi - 1) \boldsymbol{\tau}_r^b \cdot \mathbf{G}}, \quad (65)$$

$\mathbf{G}, \mathbf{H}, \mathbf{n}_r, \boldsymbol{\tau}_r^a, \chi$ are given in (21), (27), (42), (44), (51), respectively. Then, each two VFs of $\mathbf{F}_{OA}, \mathbf{G}_{OA}, \mathbf{H}_{OA}$ are orthogonal.

Proof. We compute the inner product of $\mathbf{F}_{OA}$ and $\mathbf{G}_{OA}$ as given below

$$\begin{aligned} \mathbf{F}_{OA} \cdot \mathbf{G}_{OA} &= \chi \mu \mathbf{F} \cdot \mathbf{G} + (1 - \chi)(1 - \mu) \boldsymbol{\tau}_r^b \cdot \mathbf{n}_r \\ &\quad + \chi(1 - \mu) \mathbf{F} \cdot \mathbf{n}_r + \mu(1 - \chi) \boldsymbol{\tau}_r^b \cdot \mathbf{G} \end{aligned}$$

Due to $\mathbf{F} \cdot \mathbf{G} = 0$ and $\boldsymbol{\tau}_r^b \cdot \mathbf{n}_r = 0$, the formulation of $\mathbf{F}_{OA} \cdot \mathbf{G}_{OA}$ can be reorganized as

$$\mathbf{F}_{OA} \cdot \mathbf{G}_{OA} = -\mu(\chi \mathbf{F} \cdot \mathbf{n}_r + (\chi - 1) \boldsymbol{\tau}_r^b \cdot \mathbf{G}) + \chi \mathbf{F} \cdot \mathbf{n}_r. \quad (66)$$

By substituting (65) into (66), we have $\mathbf{F}_{OA} \cdot \mathbf{G}_{OA} = 0$, indicating that the vector fields $\mathbf{F}_{OA}$ and $\mathbf{G}_{OA}$ are orthogonal. Next, we compute the cross product of $\mathbf{G}_{OA}$ and $\mathbf{F}_{OA}$.

$$\begin{aligned} \mathbf{G}_{OA} \times \mathbf{F}_{OA} &= \chi \mu \mathbf{G} \times \mathbf{F} + (1 - \chi)(1 - \mu) \mathbf{n}_r \times \boldsymbol{\tau}_r^b \\ &\quad + \mu(1 - \chi) \mathbf{G} \times \boldsymbol{\tau}_r^b + \chi(1 - \mu) \mathbf{n}_r \times \mathbf{F}. \end{aligned}$$

Owing to $\mathbf{G} \times \mathbf{F} = \mathbf{H}$, $\mathbf{n}_r \times \mathbf{F} = \boldsymbol{\tau}_r^a$ and $\mathbf{n}_r \times \boldsymbol{\tau}_r^b = \boldsymbol{\tau}_r^a$, the formulation of $\mathbf{G}_{OA} \times \mathbf{F}_{OA}$ can be further simplified to be (64), that is, $\mathbf{H}_{OA} = \mathbf{G}_{OA} \times \mathbf{F}_{OA}$, implying that $\mathbf{H}_{OA}$ is orthogonal to $\mathbf{F}_{OA}$ and $\mathbf{G}_{OA}$, respectively. ■

With the help of [Lemma 8](#), we are able to construct an auxiliary attitude matrix $\mathbf{R}_a$ according to the VFs $\mathbf{F}_{OA}, \mathbf{G}_{OA}, \mathbf{H}_{OA}$. To be more specific, by replacing the components of $\mathbf{F}, \mathbf{G}, \mathbf{H}$ in (30) with those of $\mathbf{F}_{OA}, \mathbf{G}_{OA}, \mathbf{H}_{OA}$, we can obtain an attitude matrix $\mathbf{R}_a$ for obstacle avoidance. Then, based on $\mathbf{R}_a$, the attitude tracking control input can be designed as (32). Note that in [Lemma 8](#), μ is not well defined once it holds $\chi \mathbf{F} \cdot \mathbf{n}_r + (\chi - 1) \boldsymbol{\tau}_r^b \cdot \mathbf{G} = 0$. We analyze in [Appendix B](#) this special case and show alternative methods to redefine $\mathbf{G}_{OA}$ and $\mathbf{H}_{OA}$.

Although the OA-VF $\mathbf{F}_{OA}$ (50) is proposed for one obstacle, it can be easily extended to the situation of multiple obstacles. Assume there exist M obstacles in the workspace and each one can be described by $\chi_i(\mathbf{p}; \mathbf{p}_{oi}) = 1, i = 1, \dots, M$. Then, the obstacle avoidance vector field can be specified as

$$\mathbf{F}_{OA} = \left(\prod_{i=1}^M \chi_i \right) \mathbf{F} + \sum_{i=1}^M ((1 - \chi_i) \boldsymbol{\tau}_{ri}^b), \quad (67)$$

where χ_i is the transition function of the i th obstacle, $\mathbf{F}$ is the N-VF of the robot, and $\boldsymbol{\tau}_{ri}^b$ is the tangential VF of the i th obstacle as designed in (46) and (49).

5. Cooperative motion planning

This section investigates the cooperative motion planning of multiple nonholonomic robots via VFs, where the main problem is how to achieve collision avoidance among the nonholonomic robots. Motivated by the obstacle avoidance results in the previous section, we regard other nonholonomic robots to be avoided as moving obstacles, and introduce different priorities of the nonholonomic robots to deal with the challenge of motion coupling.

By comparing (3)–(4) with (5)–(6), we observe that the setting of collision avoidance can be regarded as avoiding moving obstacles with sphere boundaries. However, the motion couplings among nonholonomic robots make the collision avoidance essentially different from the obstacle avoidance. To be more specific, the movement of the obstacle is independent of the nonholonomic robot, while in the collision avoidance, the motion of nonholonomic robots relies on each other indeed. Let us take two nonholonomic robots labeled as A and B for example. Once A enters the reactive area of B , it would intend to circumvent B for non-collision. In the meantime, B has also detected A and would take actions to avoid A as well. Such a motion coupling introduces an algebraic loop in mathematics. Thus, compared with obstacle avoidance, how to realize motion decoupling is the main challenge in collision avoidance of multiple nonholonomic robots.

In this paper, we specify a particular priority for each nonholonomic robot to deal with the motion coupling. Specifically, nonholonomic robots are labeled by $i \in \mathcal{I}_v = \{1, \dots, N\}$, and it is prescribed that the nonholonomic robot with a smaller index has higher priority. That is to say, the 1-st nonholonomic robot possesses the highest priority, while the N th is the lowest prioritized. Regarding the i th nonholonomic robot, we define the neighboring label set

$$\mathcal{N}_i = \{j \in \mathcal{I}_v : \mathbf{p}_i \in \mathcal{C}_{rj}, i \neq j\}. \quad (68)$$

The elements in $\mathcal{N}_i$ are the labels of the nonholonomic robots whose reactive area is entered by the i th nonholonomic robot. In other words, the nonholonomic robots detected by i th are collected in $\mathcal{N}_i$. Next, we select certain labels in $\mathcal{N}_i$ to define another set

$$\mathcal{N}_i^+ = \{j \in \mathcal{N}_i : j < i\}, \quad (69)$$

which collects the nonholonomic robots that are detected by the i th robot and have higher priorities than the i th robot. Then, in order to realize collision avoidance, we let the i th nonholonomic robot avoid the nonholonomic robots in $\mathcal{N}_i^+$, while ignore those in $\mathcal{N}_i / \mathcal{N}_i^+$. In this way, the motion of the high prioritized nonholonomic robots is independent from the low prioritized ones, so that the problem of collision avoidance among robots can be converted into the problem of moving obstacles avoidance.

For example, as shown in [Fig. 4](#), there are four nonholonomic robots in the workspace, and the sets $\mathcal{N}_i$ and $\mathcal{N}_i^+$ ($i \in \mathcal{I}_v = \{1, 2, 3, 4\}$) can be obtained as $\mathcal{N}_1 = \{2, 3, 4\}, \mathcal{N}_1^+ = \emptyset; \mathcal{N}_2 = \{1, 3\}, \mathcal{N}_2^+ = \{1\}; \mathcal{N}_3 = \{1, 2, 4\}, \mathcal{N}_3^+ = \{1, 2\}; \mathcal{N}_4 = \{1, 3\}, \mathcal{N}_4^+ = \{1, 3\}$. Then, the labels in $\mathcal{N}_i^+$ are the nonholonomic robots that the i th nonholonomic robot should circumvent, by regarding them as independent moving obstacles.

Therefore, similar to (42), we derive the normal vector of the sphere $\boldsymbol{\Psi}_j(\mathbf{p}; \mathbf{p}_j) = \bar{k} (r_c \leq k \leq r_d)$ as follows

$$\mathbf{n}_{\boldsymbol{\Psi}_j} = \frac{\partial \boldsymbol{\Psi}_j}{\partial x} \mathbf{e}_x + \frac{\partial \boldsymbol{\Psi}_j}{\partial y} \mathbf{e}_y + \frac{\partial \boldsymbol{\Psi}_j}{\partial z} \mathbf{e}_z, \quad (70)$$

and the time derivative of $\mathbf{n}_{\boldsymbol{\Psi}_j}$ is

$$\dot{\mathbf{n}}_{\boldsymbol{\Psi}_j} = \frac{\partial \mathbf{n}_{\boldsymbol{\Psi}_j}}{\partial \mathbf{p}} \dot{\mathbf{p}} + \frac{\partial \mathbf{n}_{\boldsymbol{\Psi}_j}}{\partial \mathbf{p}_j} \dot{\mathbf{p}}_j, \quad (71)$$

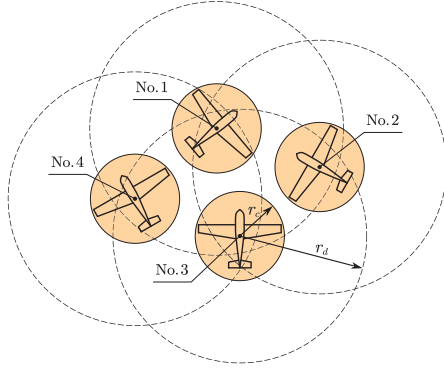


Fig. 4. Collision avoidance of multiple nonholonomic robots (taking fixed-wing UAVs for example).

Algorithm 1: Design of the CA-VF $\mathbf{F}_{CAi}$

Input: $\mathbf{p}_i$ (the present position of the i -th robot)
Output: $\mathbf{F}_{CAi}$ (the CA-VF of the i -th robot)

- 1 Determine the detection range $\mathcal{D}_i$ of the i -th robot
- 2 Detect other robots' position $\mathbf{p}_j$ within $\mathcal{D}_i$
- 3 **for each** $\mathbf{p}_j \in \mathcal{D}_i$ **do**
- 4 **if** $\mathbf{p}_i \in \mathcal{C}_{ij}$ **then**
- 5 Add j to the set $\mathcal{N}_i$
- 6 **if** $j < i$ **then**
- 7 Add j to the set $\mathcal{N}_i^+$
- 8 **end**
- 9 **end**
- 10 **end**
- 11 **for each** $j \in \mathcal{N}_i^+$ **do**
- 12 Compute the vector field $\mathbf{n}_{\Psi_j}$
- 13 **if** $\mathbf{n}_{\Psi_j} \nparallel \mathbf{F}_i$ **then**
- 14 $\boldsymbol{\tau}_{\Psi_j}^b = (\mathbf{n}_{\Psi_j} \times \mathbf{F}_i) \times \mathbf{n}_{\Psi_j}$
- 15 **else**
- 16 $\boldsymbol{\tau}_{\Psi_j}^b = \mathbf{G}_j$
- 17 **end**
- 18 Compute the transition function χ_j
- 19 **end**
- 20 Compute the CA-VF $\mathbf{F}_{CAi}$ as (72)

where $j \in \mathcal{N}_i^+$. Then, the collision avoidance VF (CA-VF) for the i th nonholonomic robot can be designed by

$$\mathbf{F}_{CAi} = \left(\prod_{j \in \mathcal{N}_i^+} \chi_j \right) \mathbf{F}_i + \sum_{j \in \mathcal{N}_i^+} ((1 - \chi_j) \boldsymbol{\tau}_{\Psi_j}^b), \quad (72)$$

where $\mathbf{F}_i$ is the N-VF of the i th nonholonomic robot in the free space, $\boldsymbol{\tau}_{\Psi_j}^b$ is the tangential VF given by

$$\boldsymbol{\tau}_{\Psi_j}^b = \begin{cases} (\mathbf{n}_{\Psi_j} \times \mathbf{F}_i) \times \mathbf{n}_{\Psi_j}, & \text{if } \mathbf{n}_{\Psi_j} \nparallel \mathbf{F}_i, \\ \mathbf{G}_i, & \text{if } \mathbf{n}_{\Psi_j} \parallel \mathbf{F}_i, \end{cases} \quad (73)$$

and χ_j is the transition function given by

$$\chi_j = \begin{cases} 0, & \mathbf{p} \in \{\Psi_j(\mathbf{p}; \mathbf{p}_j) = r_c\}, \\ S(\mathbf{p}), & \mathbf{p} \in \{r_c < \Psi_j(\mathbf{p}; \mathbf{p}_j) < r_d\}, \\ 1, & \mathbf{p} \in \{\Psi_j(\mathbf{p}; \mathbf{p}_j) = r_d\}. \end{cases} \quad (74)$$

Algorithm 1 summarizes the main procedure to design the CA-VF $\mathbf{F}_{CAi}$ of the i th nonholonomic robot.

Having designed $\mathbf{F}_{CAi}$, we can simply construct the auxiliary VFs $\mathbf{G}_{CAi}$ and $\mathbf{H}_{CAi}$ as in Lemma 8. Subsequently, based on $\mathbf{F}_{CAi}$,

$\mathbf{G}_{CAi}$, $\mathbf{H}_{CAi}$, the auxiliary attitude matrix $\mathbf{R}_a$ can be constructed as (30), and then the attitude tracking control input $\boldsymbol{\Omega}^\wedge$ can be designed accordingly by Lemma 5.

Remark 6. Although the multi-robot collision avoidance is converted to multi-obstacle avoidance by utilizing the priority-based algorithm, the CA-VF $\mathbf{F}_{CAi}$ in (72) has a difference from the OA-VF $\mathbf{F}_{OA}$ in (67). Specifically, Assumption 2 indicates that the robot only enters the reactive area of one obstacle at each time instant, so that the OA-VF $\mathbf{F}_{OA}$ in (67) naturally degenerates to (50) in the single-robot case. Nevertheless, it would be restrictive to assume that each robot always only enters the reactive area of at most one robot, because the motion of the robots cannot be predefined as the environment information. But there exist some practical rules to convert the collision avoidance among multiple robots to a pair of robots. Similar to Marchidan and Bakolas (2020), Zhao and Zhu (2022), one possible method is to let each robot avoid the nearest robot nearby, which is reasonable since the nearest robot has the largest collision risk. Of course, if more than one robots are equally the nearest, the CA-VF $\mathbf{F}_{CAi}$ possibly vanishes, leading to singular points. However, we conjecture these points are of measure zero since they rarely occur in simulations and experiments.

6. Numerical simulation results

In this section, we provide three numerical simulation examples based on fixed-wing UAVs to verify the effectiveness of the proposed motion planning algorithms.

The first example is to drive a fixed-wing UAV to the target position with the desired heading direction in an obstacle-free environment. The target position is chosen as the origin of $\mathcal{F}_{xyz}$ and the desired heading direction is specified as the positive direction of the x -axis, that is, $\mathbf{p}_d = [0 \ 0 \ 0]^T$ and $\mathbf{e}_d = \mathbf{e}_x = [1 \ 0 \ 0]^T$. The control gains in the angular and linear velocities are set to be $k_\omega = 0.5$ and $k_v = 0.1$. The simulation is carried out based on six initial conditions, including the position and orientation of the fixed-wing UAV. Fig. 5 depicts the trajectories of the fixed-wing UAV from different initial conditions, demonstrating that the fixed-wing UAV reaches the goal point with the specified heading direction. The position and orientation of the fixed-wing UAV evolving with time are shown in Fig. 6, where the orientation is parameterized as the Euler angles for the sake of illustration. Particularly, as seen in Fig. 6, the pitch and yaw angles converge to zero, indicating that the heading direction of the fixed-wing UAV points to the x -axis of $\mathcal{F}_{xyz}$. Furthermore, we remark that the convergence speed might become lower near the origin, since the norm $\|\mathbf{p}\|$ is added as a scaler into $\mathbf{F}(\mathbf{p})$ as given in (38). However, a control gain k_v is also introduced into $\mathbf{F}(\mathbf{p})$ to speed up the convergence.

In the second example, the fixed-wing UAV moves in an obstacle-cluttered environment. By following the obstacle description given in Khatib (1986), the obstacles in this simulation example are formulated by $\mathcal{T}(\mathbf{p}; \mathbf{p}_o) = \left(\frac{x-x_o}{a}\right)^{2p} + \left(\frac{y-y_o}{b}\right)^{2q} + \left(\frac{z-z_o}{c}\right)^{2r} = 1$, where p, q, r are used to describe the geometrical shape of the obstacle, and a, b, c are used to control the size of the obstacle. By choosing different p, q, r , the equation $\mathcal{T}(\mathbf{p}; \mathbf{p}_o) = 1$ represents the ellipsoid, cylinder, cone, etc. In simulation, the control gains are set to be $k_\omega = 1$ and $k_v = 0.1$. Simulation results based on three different initial conditions are given in Fig. 7, where the target position is chosen as $\mathbf{p}_d = [75 \ 30 \ 25]^T$, and the desired heading direction is specified as $\mathbf{e}_d = \mathbf{e}_x = [1 \ 0 \ 0]^T$.

The third example provides the motion planning results of seven fixed-wing UAVs with collision avoidance. As shown in Fig. 8, each fixed-wing UAV is required to reach its desired position and keep the final heading direction as $\mathbf{e}_d = [\frac{\sqrt{2}}{2} \ \frac{\sqrt{2}}{2} \ 0]^T$.

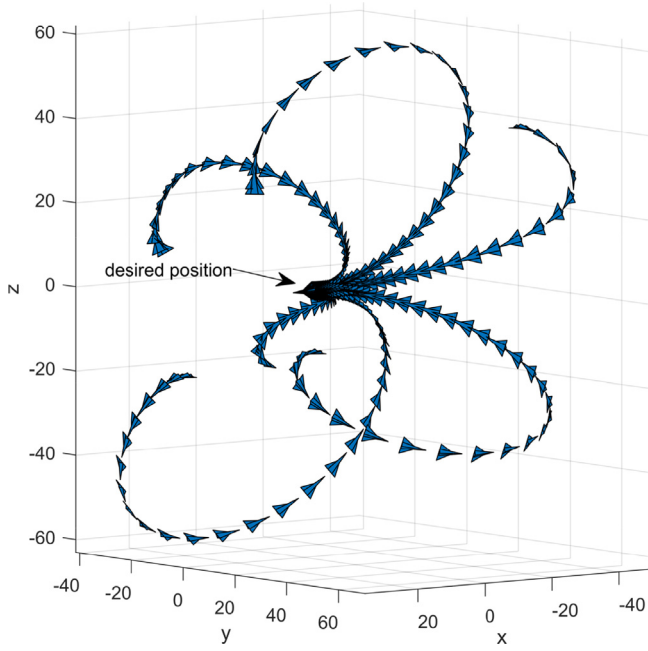


Fig. 5. Trajectories of a fixed-wing UAV moving in an obstacle-free environment (with six different initial conditions).

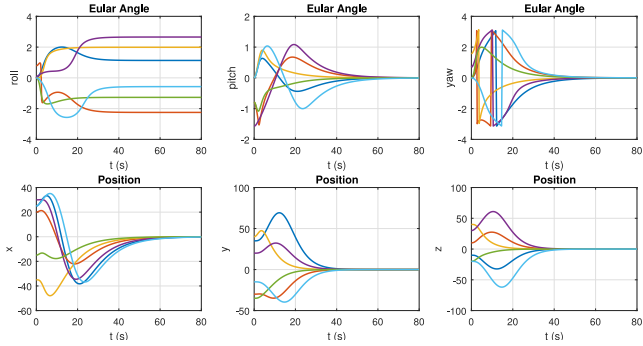


Fig. 6. Position and orientation evolution of the fixed-wing UAV.

The control gains in this simulation are chosen as $k_\omega = 0.5$ and $k_v = 0.1$. It can be observed that the fixed-wing UAVs achieve the goal of motion planning and do not collide with each other. Besides, we note that the fixed-wing UAV in the middle of the line (which is marked in dark blue) is labeled by $i = 1$, so that it has the highest priority and its movement is not influenced by others. In contrast, the fixed-wing UAV starting from the position $(0, 0, 0)$ (which is marked in dark magenta) is the lowest prioritized, and thus it has to avoid the rest of all fixed-wing UAVs.

7. Conclusion

This paper has proposed a novel VF for nonholonomic robots in 3D to solve the motion planning problem where robots are required to not only reach the specified target positions but also align with the predefined heading directions. A composite VF has been further presented to ensure collision avoidance with obstacles and other robots by guaranteeing no penetration of the no-entering area. In addition, we have proposed a priority-based algorithm to achieve motion decoupling among multiple nonholonomic robots. Future research will focus on the motion

planning problem in more practical scenarios, such as input saturation constraints, path curvature constraints and robust motion with measurement noises.

Appendix A. Proof of Lemma 2

(i) By substituting (9) into (7), the components of $\mathbf{F}$, i.e., F_x, F_y, F_z can be rewritten as

$$F_x = x^2 - r^2, F_y = 2xr \cos \theta, F_z = 2xr \sin \theta. \quad (\text{A.1})$$

According to Surana (2014), the transformation between $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ and $\{\mathbf{e}_x, \mathbf{e}_r, \mathbf{e}_\theta\}$ is given by

$$\begin{bmatrix} \mathbf{e}_x \\ \mathbf{e}_y \\ \mathbf{e}_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \mathbf{e}_x \\ \mathbf{e}_r \\ \mathbf{e}_\theta \end{bmatrix}. \quad (\text{A.2})$$

Then, based on (A.1) and (A.2) and after computation, the N-VF $\mathbf{F}$ given in (7) can be expressed in the coordinate frame $\mathcal{F}_{xr\theta}$ as (11).

(ii) This can be proved based on the results in (i). We substitute (10) into (11) and reformulate F_x, F_r, F_θ as

$$F_x = \rho^2(\cos^2 \varphi - \sin^2 \varphi), F_r = 2\rho^2 \cos \varphi \sin \varphi, F_\theta = 0. \quad (\text{A.3})$$

Referring to Surana (2014) again, we obtain the transformation between $\{\mathbf{e}_x, \mathbf{e}_r, \mathbf{e}_\theta\}$ and $\{\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{e}_\theta\}$ as

$$\begin{bmatrix} \mathbf{e}_x \\ \mathbf{e}_r \\ \mathbf{e}_\theta \end{bmatrix} = \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{e}_\rho \\ \mathbf{e}_\varphi \\ \mathbf{e}_\theta \end{bmatrix}. \quad (\text{A.4})$$

With the help of (A.3) and (A.4), the N-VF $\mathbf{F}$ given in (11) can be expressed in the coordinate frame $\mathcal{F}_{xr\theta}$ as (12).

Appendix B. Undefined cases in Lemma 8

To analyze when it holds $\chi \mathbf{F} \cdot \mathbf{n}_\gamma + (\chi - 1) \tau_\gamma^b \cdot \mathbf{G} = 0$, we firstly apply transformation to the inner products $\mathbf{F} \cdot \mathbf{n}_\gamma$ and $\tau_\gamma^b \cdot \mathbf{G}$. Note that the VFs $\mathbf{F}, \mathbf{H}, \mathbf{G}$ define an auxiliary matrix for navigation in free space, denoted by $\mathbf{R}_a^N$. Similarly, the VFs $\tau_\gamma^b, \tau_\gamma^a, \mathbf{n}_\gamma$ define an auxiliary matrix for obstacle avoidance, denoted by $\mathbf{R}_a^{OA}$. Assume that the relation between $\mathbf{R}_a^N$ and $\mathbf{R}_a^{OA}$ is given by $\mathbf{R}_a^{OA} = \bar{\mathbf{R}} \mathbf{R}_a^N$, where $\bar{\mathbf{R}} \in \text{SO}(3)$. Then, we have $\tau_\gamma^b = \bar{\mathbf{R}} \mathbf{F}$ and $\mathbf{n}_\gamma = \bar{\mathbf{R}} \mathbf{G}$. Hence, the inner products could be rewritten as $\mathbf{F} \cdot \mathbf{n}_\gamma = \mathbf{F} \cdot (\bar{\mathbf{R}} \mathbf{G})$ and $\tau_\gamma^b \cdot \mathbf{G} = (\bar{\mathbf{R}} \mathbf{F}) \cdot \mathbf{G}$, which are further expressed in vector form as $\mathbf{F}^T \bar{\mathbf{R}} \mathbf{G}$ and $\mathbf{F}^T \bar{\mathbf{R}}^T \mathbf{G}$. Therefore, the original condition $\chi \mathbf{F} \cdot \mathbf{n}_\gamma + (\chi - 1) \tau_\gamma^b \cdot \mathbf{G} = 0$ could be reformulated as

$$\mathbf{F}^T (\chi \bar{\mathbf{R}} + (\chi - 1) \bar{\mathbf{R}}^T) \mathbf{G} = 0. \quad (\text{B.1})$$

Since $\mathbf{F} \neq \mathbf{0}$ and $\mathbf{G} \neq \mathbf{0}$ (except for the trivial case at the origin), the condition (B.1) holds if and only if $\chi \bar{\mathbf{R}} = (1 - \chi) \bar{\mathbf{R}}^T$, which is simplified as

$$\chi \bar{\mathbf{R}}^2 = (1 - \chi) \mathbf{I}. \quad (\text{B.2})$$

Note that $\det(\chi \bar{\mathbf{R}}^2) = \chi$ and $\det((1 - \chi) \mathbf{I}) = 1 - \chi$, where $\det(\bar{\mathbf{R}}) = 1$ is utilized, so that the condition (B.2) holds if and only if $\chi = \frac{1}{2}$, which indicates $\bar{\mathbf{R}}^2 = \mathbf{I}$. Given the fact that $\bar{\mathbf{R}}^T \bar{\mathbf{R}} = \mathbf{I}$, it is further obtained that $\bar{\mathbf{R}} = \bar{\mathbf{R}}^T$. Thus, in Lemma 8, μ given by (65) is not well-defined if and only if $\chi = \frac{1}{2}$ and $\bar{\mathbf{R}} = \bar{\mathbf{R}}^T$. If μ is not well-defined, then depending on whether it holds $\bar{\mathbf{R}} = \mathbf{I}$, we redefine the VFs $\mathbf{G}_{OA}$ and $\mathbf{H}_{OA}$ in the following.

If $\bar{\mathbf{R}} = \mathbf{I}$, it implies that the auxiliary matrices $\mathbf{R}_a^N$ and $\mathbf{R}_a^{OA}$ are equivalent, and hence we have $\mathbf{F} = \tau_\gamma^b$ and $\mathbf{G} = \mathbf{n}_\gamma$. Thus, we could directly define $\mathbf{F}_{OA} = \mathbf{F}$ and $\mathbf{G}_{OA} = \mathbf{G}$, whose orthogonality is guaranteed by Lemma 3, and $\mathbf{H}_{OA}$ is given by

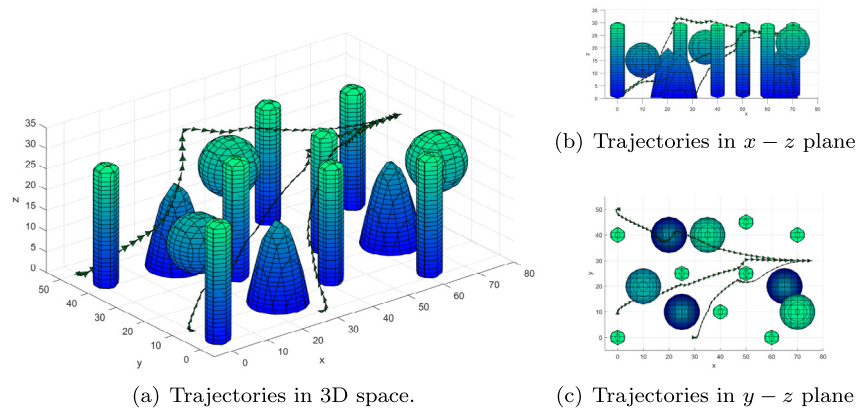


Fig. 7. Trajectories of a fixed-wing UAV moving in an obstacle-cluttered environment (three different initial conditions).

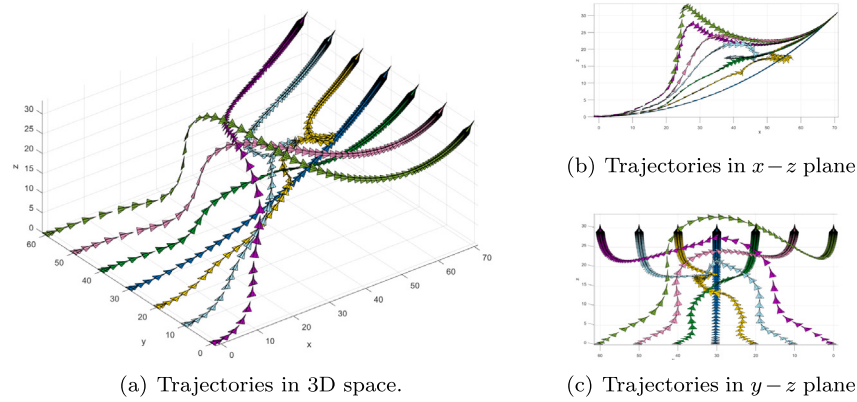


Fig. 8. Trajectories of seven fixed-wing UAVs with collision avoidance. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$\mathbf{H}_{OA} = \mathbf{G}_{OA} \times \mathbf{F}_{OA}$. If $\bar{\mathbf{R}} \neq \mathbf{I}$, we define $\mathbf{F}_{OA} = \frac{1}{2}(\mathbf{F} + \tau_{\mathcal{R}}^b)$ and $\mathbf{G}_{OA} = \frac{1}{2}(\mathbf{G} - \mathbf{n}_{\mathcal{R}})$. By substituting $\tau_{\mathcal{R}}^b = \bar{\mathbf{R}}\mathbf{F}$ and $\mathbf{n}_{\mathcal{R}} = \bar{\mathbf{R}}\mathbf{G}$, we further have $\mathbf{F}_{OA} = \frac{1}{2}(\mathbf{I} + \bar{\mathbf{R}})\mathbf{F}$ and $\mathbf{G}_{OA} = \frac{1}{2}(\mathbf{I} - \bar{\mathbf{R}})\mathbf{G}$. By utilizing $\mathbf{F} \cdot \mathbf{G} = 0$, $\bar{\mathbf{R}} = \bar{\mathbf{R}}^T$ and $\bar{\mathbf{R}}^T \bar{\mathbf{R}} = \mathbf{I}$, it is not difficult to verify that $\mathbf{G}_{OA} \cdot \mathbf{F}_{OA} = 0$, and then $\mathbf{H}_{OA} = \mathbf{G}_{OA} \times \mathbf{F}_{OA}$ is defined accordingly.

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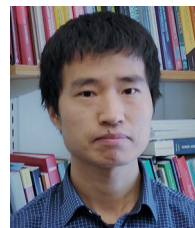


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